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Research article

AI-enabled root-cause diagnosis of water quality anomalies in continuous monitoring networks using a retrieval-augmented generation (RAG) approach

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ABSTRACT

Real-time sensor networks are widely used to monitor the quality of surface waters; however, translating detected anomalies into clear operational actions remains challenging. Deep learning (DL) models can detect deviations effectively, but they rarely provide interpretable or regulation-relevant explanations, and existing explainable artificial intelligence (XAI) tools offer limited practical value for water utilities. This study introduces a framework, Water Quality- Retrieval Augmented Generation (WQ-RAG), that integrates DL-based anomaly detection with RAG to enable automated, regulation-grounded root-cause diagnosis of water quality events. The framework was evaluated using high-frequency monitoring data from 2021 to 2024 at four geographically and hydrogeochemically diverse river monitoring stations operated by the U.S. Geological Survey (USGS). Among four anomaly detection models, a Long Short-Term Memory (LSTM) autoencoder achieved the highest F1-scores at three stations (0.799-0.841), while the Patch Time Series Transformer (PatchTST) showed the strongest discrimination, measured by the area under the receiver operating characteristic (ROC) curve (AUC-ROC: 0.860-0.906). Adaptive, station-specific thresholding improved detection performance by up to 61% at low-variance source water. Using a curated knowledge base of 25 regulatory and technical documents, RAG-grounded explanations generated by a large language model (LLM) (Llama-3-8B) achieved a mean quality score of 0.840 ± 0.071 across completeness, regulatory accuracy, and actionability. An ablation analysis showed that retrieval grounding increased regulatory accuracy 3.8-fold, from 0.20 to 0.76, confirming that LLMs lack intrinsic knowledge of water quality standards. These results demonstrate that retrieval grounding is essential for converting anomaly alerts into reliable, regulation-aware diagnoses that support timely and effective water quality management.

1. Introduction

Continuous, real-time monitoring of physicochemical parameters across geographically distributed sensor networks is critical for assessing water quality, supporting resource management, and enabling the early detection of environmental disturbances and contamination events (Storey et al., 2011; Yaroshenko et al., 2020). The United States Geological Survey (USGS) operates more than 13,500 real-time monitoring locations across the United States, including stream, lake, reservoir, precipitation, water-quality, and groundwater monitoring locations (USGS, 2026a). Continuous surface water-quality monitors within this network typically record water temperature, specific conductance (SC), dissolved oxygen (DO), pH, and turbidity at regular intervals (Wagner et al., 2006). Collectively, these physicochemical parameters provide a comprehensive characterization of water bodies and are widely used to assess water quality, identify contamination, monitor nutrient dynamics, and evaluate ecosystem condition (Chidiac et al., 2023; Zhi et al., 2024). However, the volume and velocity of these high-frequency data streams exceed the capability of manual expert review, in which the checking, labelling, and correction of each observation becomes unfeasible at scale (Leigh et al., 2019). This creates a significant need for automated systems that can not only detect

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0301-4797/© 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

anomalous patterns but also diagnose their probable causes and recommend regulatory-compliant response actions.

Building on this need for automated analysis, data-driven modelling of surface water quality has evolved from classical machine-learning (ML) and hybrid metaheuristic-optimized models toward deep learning (DL) architectures that learn directly from high-frequency sensor data streams (Pham et al., 2021; Wai et al., 2022). DL models, including continental-scale frameworks, have been applied to predict individual water quality parameters such as DO (Zhi et al., 2021), and representation-learning approaches now enable knowledge transfer and cross-basin prediction under data-scarce conditions (Zheng et al., 2025). In parallel, anomaly detection in water quality monitoring has progressed from statistical outlier methods toward DL architectures capable of modelling the complex temporal dependencies in multivariate sensor data (Jeong et al., 2024; Santos-Fernandez et al., 2024; Zhi et al., 2024). These developments build upon broader advances in DL for time-series anomaly detection, for which several taxonomies have been proposed based on anomaly-detection strategies and DL model architecture. Among DL architectures, the Long Short-Term Memory (LSTM) network has become a foundational model for water-quality time-series modelling because of its ability to capture long-range temporal dependencies through gated memory-cell mechanisms. This capability enables the learning of sequential and seasonally driven dynamics in key water-quality parameters, including total nitrogen (TN) and DO (Gao et al., 2023), turbidity (Song and Zhang, 2020), as well as the interactions between DO and its major driving factors, such as temperature, pH, and conductivity (Liang et al., 2024).

Currently, transformer architectures have advanced multivariate time-series modelling through self-attention mechanisms that capture complex inter-parameter dependencies. The Patch-based Time Series Transformer (PatchTST) model, which segments time series into subseries-level patches, demonstrated improved long-term forecasting performance compared with earlier transformer-based architectures across multiple benchmark datasets (Nie et al., 2022). However, subsequent work has demonstrated that carefully designed linear models can achieve performance comparable to transformer-based approaches on several forecasting tasks (Zeng et al., 2023). Similarly, other methods such as LSTM-AutoEncoder (AE) (Malhotra et al., 2016), Anomaly Transformer (Xu et al., 2022), and TranAD (Tuli et al., 2022) achieve high anomaly detection accuracy on sensor data. These architectures have since been adapted for water quality monitoring and to detect anomalies in real-time river and sensor-network water quality data (El-Shafeiy et al., 2023; Ramya et al., 2025; Wang et al., 2023; Zhao et al., 2020).

Despite substantial advances in anomaly-detection methodologies, a fundamental limitation remains: most approaches can identify anomalous observations but provide limited insight into their underlying causes. Even models demonstrating high detection performance in water-quality monitoring applications primarily focus on anomaly identification rather than explaining anomaly sources, evaluating their regulatory implications, or informing corrective actions (Santos-Fernandez et al., 2024). This lack of interpretability remains a significant challenge for the broader adoption of DL approaches in water-quality monitoring and management (Zhi et al., 2024).

The growing use of DL in hydrological and water-quality modelling has highlighted the need for explainable artificial intelligence (XAI) approaches to improve model interpretability and transparency (Başağaoğlu et al., 2022). Post-hoc attribution methods, most notably Shapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME), have been widely applied to interpret water quality prediction models (Infant et al., 2025; Makumbura et al., 2024; Nallakaruppan et al., 2024; Park et al., 2022; Wang et al., 2025). However, these methods produce feature-importance rankings rather than operational diagnoses, and their explanations can be unstable; small perturbations to the model or input can yield different attribution (Panigrahi et al., 2025; Robert et al., 2024). Importantly, water utility operators need explanations that support decision-making, including what an anomaly signifies and the actions it warrants, rather than statistical attributions identifying the most influential input variables (Maltbie et al., 2021).

In the era of generative AI, Large Language Models (LLMs) offer a fundamentally different capability: the generation of natural-language diagnoses that integrate causal reasoning, regulatory knowledge, and actionable recommendations (Li et al., 2026). However, the application of LLMs to water quality diagnosis remains challenging due to issues such as hallucinations, factual inaccuracies, and uncertainties regarding their reliability in domain-specific environmental applications (Arslan, 2026; Foroumandi et al., 2023; Rillig et al., 2023). Crucially, LLMs exhibit hallucinations that cannot be fully eliminated through training or scale alone (Farquhar et al., 2024; Xu et al., 2025). These limitations indicate that general-purpose LLMs lack sufficient parametric knowledge of the specialized, jurisdiction-dependent regulatory frameworks governing water quality. Standards such as the U.S. EPA National Primary Drinking Water Regulations (NPDWR) (US EPA, 2026a), state-level water quality standards (Oregon Department of Environmental Quality, 2026), and species-specific aquatic life criteria constitute highly specialized knowledge that is underrepresented in general pre-training data.

Retrieval-Augmented Generation (RAG), introduced by Lewis et al. (2020), augments LLMs with an external retrieval mechanism that supplies relevant documents at inference time. This approach grounds generated responses in retrieved evidence, thereby improving factual accuracy and interpretability while facilitating efficient knowledge updates without requiring extensive retraining or fine-tuning of the underlying model. Since its introduction, the paradigm has developed rapidly along several complementary directions. Houria and Faiçal (2025) proposed a four-stage taxonomy for RAG optimization comprising indexing, retrieval, augmentation and generation, and provided a comprehensive review of techniques including chunking strategies, embedding methods, query expansion, context compression, and prompt engineering (Houria and Faiçal, 2025). Beyond pipeline-level optimization, Yang et al. (2025) proposed a multi-round retrieval framework that learns when additional retrieval is necessary, whereas Sarthi et al. (2023) developed a hierarchical approach that improves reasoning over long documents through recursive summarization (Yang et al., 2025; Sarthi et al., 2023).

The effectiveness of retrieval augmentation has been consistently demonstrated across knowledge-intensive scientific disciplines. By grounding generated responses in curated domain literature, RAG systems have been shown to improve factual accuracy, enhance traceability, and mitigate hallucinations in applications spanning chemistry, medicine, and materials science (Zhong et al., 2025; Xiong et al., 2024; Krotkov et al., 2025). This capability is particularly relevant to regulatory compliance assessment in water quality monitoring, where decisions depend on accurate interpretation of specialized, frequently updated, and evidence-based regulatory information.

Compared with other knowledge-intensive domains, the use of LLMs and RAG frameworks in the water sector is still developing, with only a small but growing body of literature addressing water-related applications. Xu et al. (2025) identified RAG as a promising knowledge-augmentation strategy for water and wastewater management while noting that conventional text-similarity retrieval may produce incomplete results (Xu et al., 2025). Water-specific foundation models, including OceanGPT (Bi et al., 2024) and WaterGPT (Ren et al., 2024), have also begun to appear in the literature. Related developments include the use of generative AI for water utility operations (Sela et al., 2025) and an RAG-LLM framework for aquaculture management (Danvirutai et al., 2025). However, to the best of our knowledge, no published study has integrated a RAG framework with unsupervised DL-based anomaly detection to enable automated root-cause diagnosis of water quality anomalies from continuous monitoring data.

Despite recent advances, existing approaches address only parts of

the water quality monitoring problem and anomaly detection. DL models can detect anomalies but offer limited diagnostic capability, while LLMs can generate explanations only when supported by reliable domain knowledge. Post-hoc explainability methods provide feature attributions but often lack operational relevance. As a result, there is a need for an integrated framework that combines anomaly detection, root-cause diagnosis, and regulatory knowledge grounding within a single decision-support system.

To address this gap, we propose a WQ-RAG framework that integrates unsupervised DL-based anomaly detection with RAG to enable automated root-cause diagnosis of water quality anomalies directly from continuous monitoring data. The framework was developed and evaluated using high-frequency observations from four hydrogeochemically diverse USGS monitoring stations. Following standardized preprocessing and windowing, four anomaly detection models were assessed using an adaptive, station-specific thresholding strategy. A domain-specific knowledge base comprising regulatory documents, technical literature, and case studies was then constructed and integrated into a RAG pipeline to generate structured diagnostic explanations. Finally, the generated explanations were evaluated across three operational dimensions, and the contributions of individual framework components were quantified through an ablation study and a comparative assessment of two open-source LLMs.

2. Materials and methods

2.1. Study area and data collection

To evaluate the generalizability of the WQ-RAG framework across diverse hydrogeochemical conditions, four continuous water quality monitoring stations operated by the USGS were selected, spanning four distinct geographic regions of the US, as presented in Table 1.

Raw data, including the five core surface water quality parameters, were retrieved from all four stations through the USGS National Water Information System (NWIS) Representational State Transfer- Application Programming Interface (REST-API) (USGS, 2026f).

Fig. 1 shows the geospatial distribution of the four USGS stations used in this study, plotted on a map of the continental US.

Stations were selected based on four criteria: (i) availability of all five core parameters listed in Table 2; (ii) a consistent 15 min sampling interval throughout the study period; (iii) data completeness exceeding 80%; and (iv) representation of diverse hydrological regimes and regulatory contexts.

The four stations represent a diverse range of environmental conditions. The Potomac River (USGS, 2026b) is a mid-Atlantic urban river affected by combined sewer overflows. The White River (USGS, 2026c) at Hazleton drains central Indiana and is dominated by row-crop agriculture on glacial-till soils. The Clackamas River (USGS, 2026d) is a Pacific Northwest source-water river of Cascade volcanic geology, subject to post-wildfire sediment loading. The Sacramento River (USGS, 2026e) is a tidally influenced agricultural river in California's Central Valley. Together, these sites span a wide range of hydrological regimes, land uses, and baseline water chemistry, from pristine, low-conductivity mountain rivers to high-conductivity agricultural streams. These five parameters (temperature, SC, DO, pH, and turbidity) serve as the multivariate input features for all detection models, yielding an input tensor $x \in \mathbb{R}^{96_x5}$ per sliding window.

Table 1
Hydrogeochemical characteristics of the selected USGS monitoring stations (2021-2024).

USGS ID	Station Name	Region	Raw Rows	Clean Rows	SC (µS/cm)
01,646,500 (USGS, 2026b)	Potomac River, DC	East Coast	139,523	134,391	200-600
03,351,000 (USGS, 2026c)	White River, IN	Midwest	139,584	114,702	400-1200
14,211,010 (USGS, 2026d)	Clackamas River, OR	Pacific NW	139,851	~135,000	30-65
11,447,650 (USGS, 2026e)	Sacramento River, CA	West Coast	140,240	~138,000	100-300

2.2. Proposed approach

This section introduces the WQ-RAG framework, as shown in Fig. 2, developed in this study for automated root-cause diagnosis of water quality anomalies. The proposed WQ-RAG framework is a modular pipeline that combines unsupervised time-series anomaly detection with RAG. Rather than only flagging that an anomaly has occurred, the framework identifies its probable cause, assesses the relevant regulatory context, and recommends response actions. The directional logic of Fig. 2 can be traced along a single forward data path augmented by two grounding routes: anomaly detection and the RAG system. In Stage I, raw 15-min observations of the five core water quality parameters are retrieved from each station through the USGS NWIS REST-API. The data then undergo a series of preprocessing steps, including sentinel value removal, physical bounds filtering, gap handling, station-specific z-score normalization, and sliding window segmentation, producing standardized 96 step windows for subsequent analysis. These windows feed Stage II, where reconstruction-based detectors and adaptive per-station thresholding convert continuous reconstruction error into binary anomaly flags together with a severity score and the parameter-level z-score deviations. At this stage, the workflow splits into two parallel paths. Along the first grounding route, the three most-deviated parameters are composed into a natural-language retrieval query that enters Stage III, where the Maximal Marginal Relevance (MMR) retriever returns six complementary chunks from the regulatory knowledge base indexed in ChromaDB (Chroma, 2025), an open-source vector database that stores document embeddings and supports efficient similarity search. Along the second grounding pathway, the numerical anomaly context, including parameter values, deviation scores, severity level, and station identity, is forwarded directly to Stage IV. At this stage, the retrieved regulatory information and anomaly-specific context are integrated through a structured prompt and processed by the local LLM to generate a diagnostic explanation. The resulting output is subsequently assessed by the evaluation module.

2.3. Data preprocessing

Raw sensor data from all four stations were processed through a single standardized pipeline. USGS sentinel values (−999,999) were replaced with NaN markers, and physically implausible readings were removed through bound filtering (e.g. temperature outside −5 to 45 °C, pH outside 2 to 14, DO outside 0 to 25 mg/L, and turbidity outside 0 to 5000 FNU). Short gaps of 1 h or less were filled by forward interpolation, and longer gaps were excluded. Each station's series was split chronologically into training (70%), validation (15%), and test (15%) partitions to preserve temporal ordering and prevent data leakage. Each parameter was then standardized to zero mean and unit variance using training-set statistics. The normalized series was segmented into overlapping 96-step windows of 24 h with a 24-step (6-h) stride, capturing full diurnal cycles.

Because manual labelling is unfeasible at the scale of high-frequency monitoring data, ground-truth labels were generated independently of the detection models using a statistical exceedance criterion. A window was labelled anomalous if any parameter deviated by more than $\pm 3\sigma$ from the training-period mean, with all thresholds computed on the training partition alone to prevent leakage. As a further check, flagged windows were compared against separately identified extreme events (|

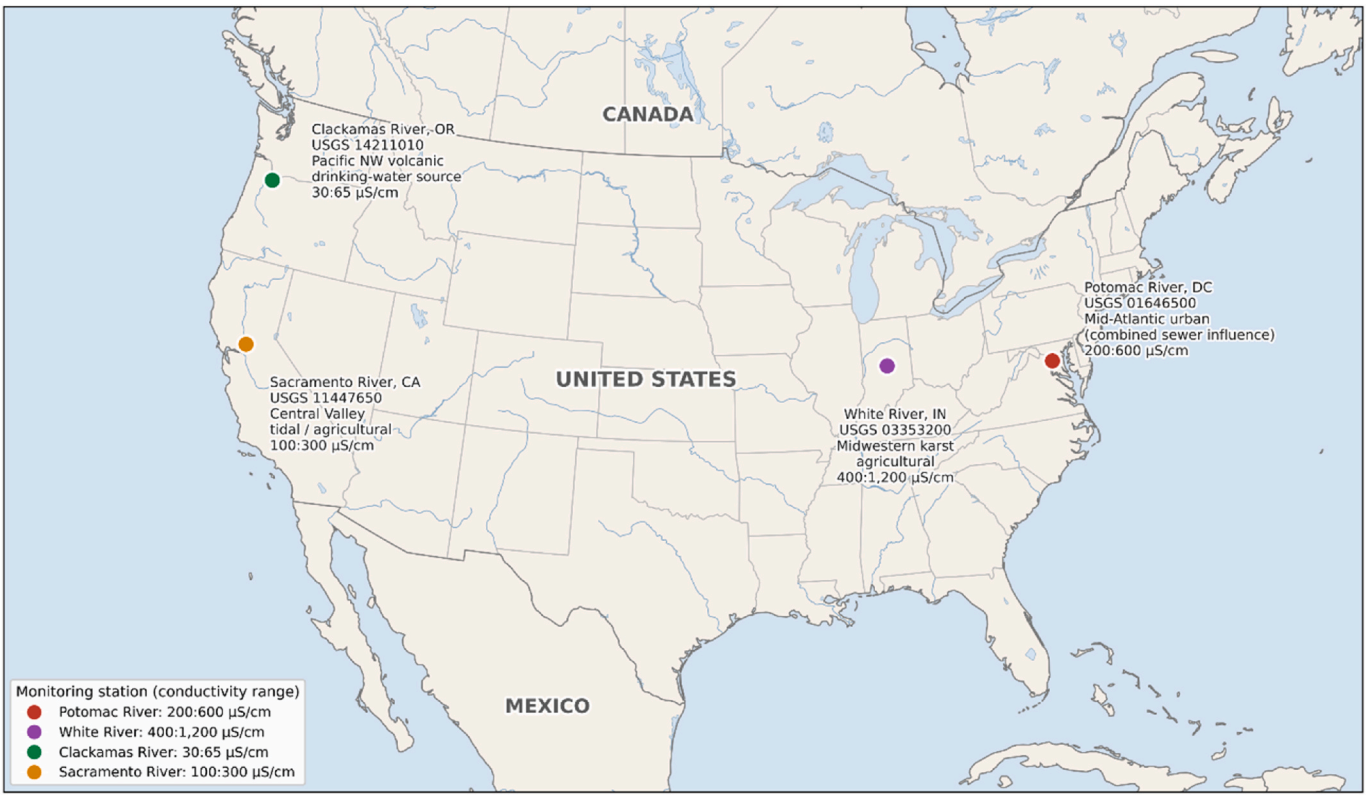


Fig. 1. Locations of the four USGS continuous water-quality monitoring stations across the United States, color-coded by hydrological regime and conductivity range.

Table 2
Input water quality parameters across all four USGS monitoring stations, with corresponding USGS parameter codes and measurement units.

S.No	Parameter	USGS Code	Unit
1	Temperature (T)	00,010	°C
2	SC	00,095	µS/cm
3	DO	00,300	mg/L
4	pH	00,400	—
5	Turbidity	63,680	FNU

$z| > 4\sigma$) and known hydrological causes, confirming that they corresponded to real changes in water quality rather than statistical artefacts.

2.4. Anomaly detection

The anomaly detection module uses an unsupervised, reconstruction-based approach in which each model is trained only on normal water quality patterns and flags anomalies through elevated reconstruction errors. This design suits water quality monitoring, where labelled anomaly datasets are scarce and the full range of contamination events cannot be defined in advance (Sakurada and Yairi, 2014). Four models were evaluated: two deep-learning architectures (PatchTST and LSTM-AE) and two conventional baselines (Isolation Forest (IF) and One-Class Support Vector Machine (SVM)). The PatchTST architecture adapts into an autoencoder, dividing each 96-step, 5-parameter window into six 4-h patches, projecting them into a 128-dimensional embedding with an encoder of three Transformer layers, and reconstructing the input through a 64-dimensional bottleneck and a two-layer decoder. An LSTM-AE (Malhotra et al., 2016) serves as a recurrent baseline, compressing each window into a fixed-length representation and reconstructing it through a symmetric decoder. Two non-temporal baselines complete the set: IF (Liu et al., 2008), using 200 isolation trees, and One-Class SVM (Cortes and Vapnik, 1995; Schölkopf et al., 2001), using an RBF-kernel decision boundary. All hyperparameters were selected on the validation set, and both DL models were trained identically using the AdamW optimizer (Loshchilov and Hutter, 2018) with early stopping, so that performance differences reflect architectural capability rather than training configuration.

Continuous reconstruction errors were converted into binary anomaly labels using a threshold θ. Rather than applying a single fixed rule, θ was optimized for each station on the validation set. Candidate thresholds were drawn from statistical rules ($\theta = \mu(e) + \alpha\cdot\sigma(e)$, $\alpha \in \{1.0–4.0\}$) and data percentiles (P90–P99). The threshold maximizing the validation F1-score was selected, subject to a minimum precision of 0.65. This calibration was applied identically at every station.

2.5. RAG

The RAG pipeline grounds each generated explanation in authoritative domain knowledge retrieved at inference time. Instead of relying on the language model's internal knowledge alone, the pipeline retrieves relevant regulatory and technical document information for each detected anomaly and supplies it to the model as context. This ensures that explanations cite the correct water quality standards and thresholds and allows the knowledge base to be updated without retraining the model. The RAG pipeline comprises the following four main components, including knowledge base composition, document processing and indexing, retrieval, and diagnostic generation.

2.5.1. Knowledge base composition

The knowledge base provides the domain knowledge that grounds each generated explanation, so that the system's diagnoses rest on real regulatory limits, established monitoring practice, and past events rather than on the language model's own assumptions. It comprises 25 documents, processed into text chunks and organized into four categories, each serving a distinct role in the diagnostic process.

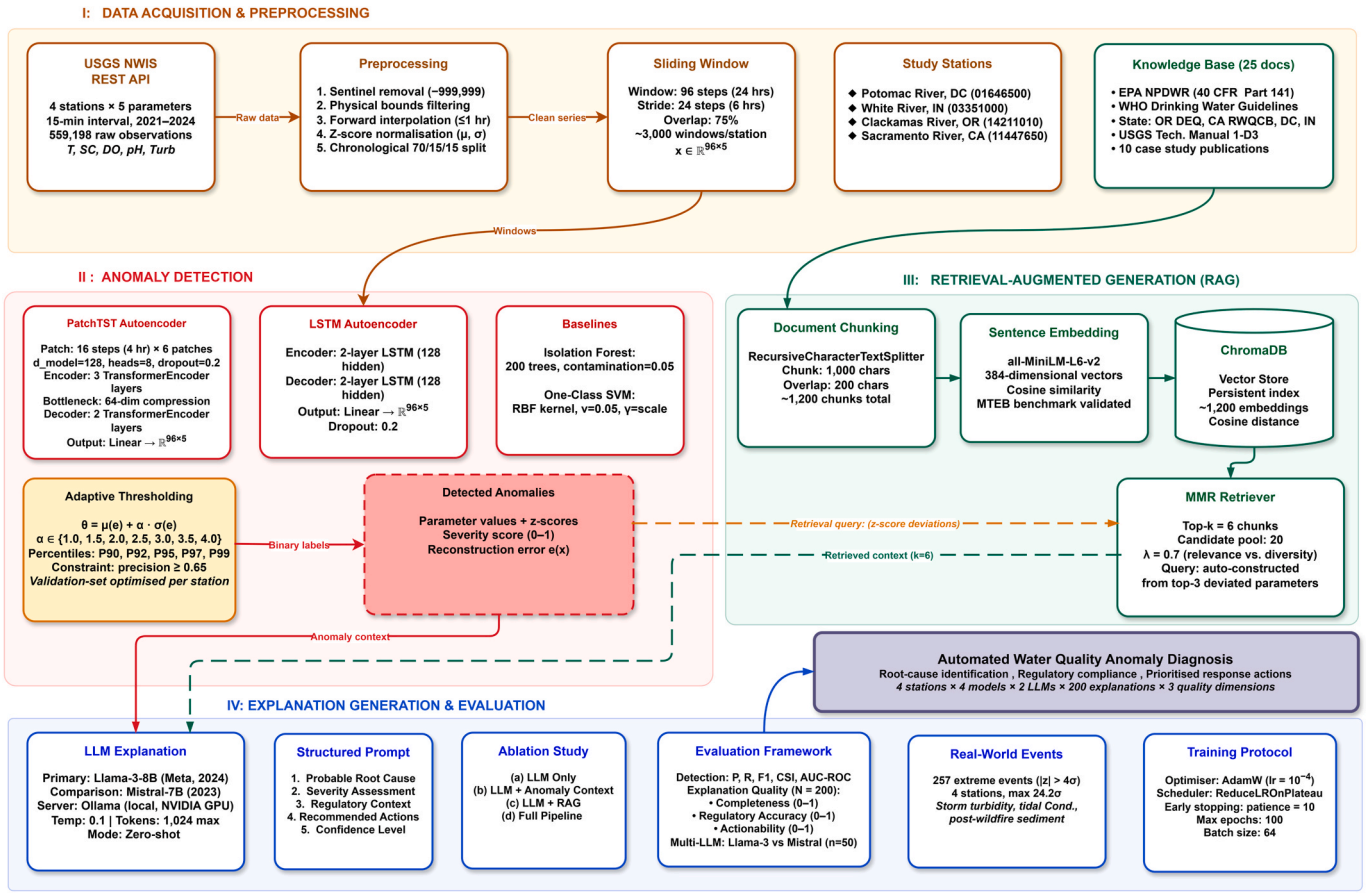


Fig. 2. System architecture of the WQ-RAG framework comprising four interconnected stages: (I) data acquisition and preprocessing from four USGS continuous monitoring stations, (II) unsupervised anomaly detection with adaptive per-station thresholding, (III) RAG grounded in a curated regulatory knowledge base, and (IV) LLM-based explanation generation and multi-dimensional quality evaluation.

1. Regulatory standards define the compliance thresholds against which anomalies are assessed. These include the U.S. EPA National Primary and Secondary Drinking Water Regulations (US EPA, 2026a, 2026b) and the WHO Guidelines for Drinking-Water Quality (WHO, 2022). By incorporating these regulatory benchmarks, the system can compare flagged readings against established legal limits and accurately determine whether a measurement constitutes a regulatory exceedance, thereby supporting reliable compliance assessment and informed decision-making.
2. Technical guides describe how the monitoring data are collected, interpreted, and evaluated against regulatory benchmarks. These include the USGS standard procedures for continuous water-quality monitors (Wagner et al., 2006) and the complete USGS parameter-code reference used to identify each measured constituent (USGS, 2026g). As the four monitored rivers serve as source waters for downstream drinking-water supplies, they are evaluated against drinking-water benchmarks, which represent the thresholds most relevant to protecting public health. Regulatory benchmarks are drawn from the EPA National Primary and Secondary Drinking Water Regulations (US EPA, 2025), the US EPA, 2018 Drinking Water Standards and Health Advisories (US EPA, 2018), and EPA drinking-water monitoring and compliance guidance (US EPA, 2026c). Together, these references support the correct interpretation of monitoring data and provide the regulatory and technical benchmarks against which measured water-quality parameters are assessed.
3. Case studies (Bonet et al., 2026; El-Shafeiy et al., 2023; Hou et al., 2013; Leigh et al., 2019; Liao et al., 2024; Liu et al., 2020; Mao et al., 2017; Perelman et al., 2012; Santos-Fernandez et al., 2024; Wang et al., 2023) provide documented evidence of water-quality anomalies, associated environmental conditions, and identified causes. These studies serve as valuable reference sources that enable the system to recognize recurring patterns, support anomaly interpretation, and infer plausible causes for newly observed events based on similarities to previously reported cases.
4. Station metadata provide site-specific context, including upstream land use, known environmental stressors, and seasonal patterns. This information allows the system to account for local environmental conditions when interpreting water-quality observations, resulting in more accurate and context-aware anomaly assessments.

2.5.2. *Document processing and vector indexing*

Documents were segmented into 1000-character chunks with 200-character overlap using LangChain's (Chase, 2022) recursive character text splitter. The preservation of contextual continuity across chunk boundaries is critical for regulatory documents, where threshold values and applicability conditions often span paragraphs. Each chunk was embedded into a 384-dimensional vector using the all- Mini Language Model (MiniLM)-L6-v2 (Wang et al., 2020) sentence transformer (Reimers and Gurevych, 2019), selected for its high performance on the Massive Text Embedding Benchmark (MTEB) while maintaining computational efficiency on consumer hardware. The resulting ~1200 embeddings were indexed in ChromaDB using cosine similarity, enabling sub-second retrieval.

2.5.3. *Retrieval strategy*

At inference time, a natural language query is automatically constructed from each detected anomaly by identifying the three most

deviated parameters and their direction. This query retrieves k = 6 chunks via MMR (Carbonell and Goldstein, 1998) with diversity parameter λ = 0.7 and a candidate pool of MMR that balances relevance to the query against diversity among the selected chunks. This prevents the retrieval of several near-identical passages and instead returns complementary information spanning the regulatory standards, technical guides, and domain references needed to explain the anomaly.

2.6. Diagnostic reasoning and evaluation

2.6.1. Explanation generation and ablation design

The explanation module receives the anomaly context, including parameter values, z-score deviations (distance from the mean in standard deviation units), severity, and station identity, together with six retrieved regulatory chunks. A structured prompt then guides the model to generate five sections that include probable root cause, severity assessment, regulatory context, prioritized recommended actions, and confidence level (a representative output is provided in **Supplementary Data S1**). The primary model is Llama-3-8B (Llama3, 2024), run locally via Ollama (Marcondes et al., 2025) for offline operation, and Mistral-7B (Mistral, 2026) serves as a comparison under identical settings. No fine-tuning is applied, so diagnostic capability derives entirely from the knowledge base and prompt, ensuring transferability across backends.

In the ablation study design, Clackamas River Station was selected as a test case due to its low-variance water chemistry and stringent regulatory requirements. Four configurations were selected for the ablation study: (1) LLM Only, with no anomaly data or retrieved knowledge; (2) LLM + Anomaly Context, using sensor data without retrieval; (3) LLM + RAG, using retrieved regulatory documents without anomaly-specific information; and (4) Full Pipeline, combining anomaly context and RAG to represent the complete framework.

2.6.2. Evaluation framework

Multi-LLM comparison: The comparison between the two LLMs was performed on 50 anomalies from the Clackamas River station, the same set used in the ablation study, so that the two analyses remain directly comparable. Both Llama-3-8B and Mistral-7B were run under identical conditions, using the same retrieved document sets, prompt templates, and RAG configuration. This ensured that any difference in performance reflected the reasoning capability of the model itself, rather than differences in the input data or experimental setup. The magnitude of the difference between the two models was quantified using Cohen's *d*, a standard measure of effect size (Cohen, 1988).

Detection performance metrics: Performance was evaluated using Precision, Recall, F1-score, and the Critical Success Index (CSI) as shown in equation (1):

$$CSI = TP \,/\, (TP + FP + FN) \tag{1}$$

The TP, FP, and FN represent the numbers of true positives, false positives, and false negatives, and CSI penalizes both false alarms and missed detections, which are equally essential for water monitoring, where both carry operational consequences. Area Under the Receiver Operating Characteristic (ROC) Curve (AUC-ROC) was computed for models with continuous scores. Complete confusion matrices are reported for all model station combinations.

Explanation quality and extreme-event validation: Explanation quality was assessed for all 200 generated explanations (50 per station) across three dimensions using automated text analysis. Completeness measures the fraction of the five required report sections that are present. Regulatory accuracy measures the fraction of five checks satisfied, namely the presence of an EPA (US EPA, 2018; US EPA, 2026c; US EPA, 2025) numeric thresholds with units, maximum contaminant level values, named regulatory frameworks, and specific guideline documents. Actionability measures the fraction of four checks satisfied, namely the presence of immediate actions, specific operational instructions, a follow-up timeframe, and reference to the relevant authorities. The overall quality score is the mean value of these three dimensions. Finally, to validate the framework against real events, naturally occurring extreme events were identified whenever any parameter deviated by more than four standard deviations from the training-period mean ($|z| > 4\sigma$). This stricter threshold isolates clear real-world anomalies, such as storm-driven turbidity spikes, tidal conductivity excursions, and post-wildfire sediment mobilization, creating an independent validation set (Liao et al., 2024).

2.7. Implementation details

The framework was implemented in Python 3.12 using PyTorch with Compute Unified Device Architecture (CUDA) acceleration, LangChain for RAG orchestration, ChromaDB for vector storage, and Ollama (2023) for local LLM inference. The two conventional baseline models, IF and One-Class SVM, were implemented using the scikit-learn ML library (Park et al., 2022). The complete pipeline covered preprocessing across four stations, training 16 detection models (4 architectures × 4 stations), constructing the vector index, and generating 200 RAG-grounded explanations with 50 Mistral comparisons. It also included the four-configuration ablation study and automated evaluation. All components were executed on a single NVIDIA GPU (16 GB Video Random Access Memory (VRAM)) with 16 GB system Random Access Memory (RAM).

3. Results

This section presents the experimental results of the WQ-RAG framework evaluated across four geographically diverse USGS monitoring stations spanning the East Coast (Potomac, DC), Midwest (White, IN), Pacific Northwest (Clackamas, OR), and West Coast (Sacramento, CA).

3.1. Anomaly detection performance

Table 3 reports on the anomaly detection performance of the four models across all monitoring stations. Performance is measured using metrics such as precision, recall, F1-score, the CSI, AUC-ROC, and the underlying confusion matrix. The models adopt distinctly different detection strategies, and each produces its own trade-off between sensitivity and false alarm control.

IF was the most conservative of the four models. It achieved the highest precision at three stations, namely Potomac (0.847), Sacramento (0.752), and Clackamas (0.767), but this came at the cost of much lower recall (0.328 to 0.701) and a large number of missed events. At the Potomac station, for instance, it detected only 150 of 458 anomalies (32.8%) and missed 308. PatchTST and OC-SVM occupied the opposite end of the performance spectrum. PatchTST achieved perfect recall (1.000) at every station along with the highest AUC-ROC values (0.860 to 0.906), and OC-SVM maintained near-perfect recall (0.919 to 0.989). Both models achieved this high sensitivity at the expense of precision (0.608 to 0.672). However, this improved detection capability came at the expense of a markedly elevated false-positive rate.

Since the operational monitoring requires a balance between missed detections and false alarms, the composite F1 and CSI metrics are the most informative. On these measures, the LSTM-AE was the strongest overall performer. It achieved the highest F1-score at three stations, namely Potomac (0.819), Sacramento (0.841), and Clackamas (0.799), by pairing high recall (0.882 to 0.985) with far fewer false positives than PatchTST. At Potomac, false positives fell from 295 to 124, and at Sacramento, it was from 294 to 166. At the White station, OC-SVM achieved the highest F1-score (0.789), just ahead of PatchTST (0.785). This is consistent with that station's greater SC variability and more distinct anomaly signatures. The CSI, which penalizes false alarms and missed detections together, tells the same narrative. The LSTM-AE achieved the highest CSI at every station (0.610 to 0.726), consistently ahead of

Table 3
Anomaly detection performance across four USGS monitoring stations. AUC-ROC is reported only for models with continuous anomaly scores.

Station	Model	Precision	Recall	F1	CSI	TP	FP	FN	AUC
Potomac, DC	PatchTST	0.608	1.000	0.756	0.608	458	295	0	0.906
	LSTM-AE	0.765	0.882	0.819	0.694	404	124	54	0.864
	IF	0.847	0.328	0.472	0.309	150	27	308	—
	OC-SVM	0.617	0.989	0.760	0.613	453	281	5	—
White River, IN	PatchTST	0.646	1.000	0.785	0.646	217	119	0	0.896
	LSTM-AE	0.704	0.820	0.757	0.610	178	75	39	0.695
	IF	0.712	0.581	0.640	0.470	126	51	91	—
	OC-SVM	0.672	0.954	0.789	0.651	207	101	10	—
Sacramento, CA	PatchTST	0.619	1.000	0.764	0.619	477	294	0	0.860
	LSTM-AE	0.738	0.979	0.841	0.726	467	166	10	0.870
	IF	0.752	0.501	0.601	0.430	239	79	238	—
	OC-SVM	0.633	0.979	0.769	0.624	467	271	10	—
Clackamas, OR	PatchTST	0.629	1.000	0.773	0.629	479	282	0	0.892
	LSTM-AE	0.672	0.985	0.799	0.666	472	230	7	0.953
	IF	0.767	0.701	0.733	0.578	336	102	143	—
	OC-SVM	0.653	0.919	0.763	0.617	440	234	39	—

PatchTST (0.608 to 0.646), OC-SVM (0.613 to 0.651), and IF (0.309 to 0.578). This robustness was made possible by the adaptive thresholding strategy. Its value was clearest at the low-variance Clackamas drinking-water source (Carpenter, 2003), where PatchTST's F1-score rose from 0.479 under a fixed threshold to 0.773 with adaptive thresholding, an improvement of 61%.

Fig. 3 summarizes the distribution of precision, recall, F1-score, and accuracy across the monitoring stations (Fig. 3a–d). Fig. 3a shows precision, the proportion of flagged windows that are true anomalies. The IF leads, reaching 0.85 at Potomac, while PatchTST and OC-SVM cluster at moderate values of roughly 0.61 to 0.67, and the LSTM-AE sits between them at 0.66 to 0.78. Fig. 3b shows recall, and the ranking reverses. PatchTST reaches perfect recall (1.00), and OC-SVM stays high at 0.92 to 0.99, whereas IF drops to as low as 0.33. This confirms that its strong precision comes at the cost of many missed events, which is a serious drawback in water-quality monitoring, where an undetected contamination event can have severe consequences. Taken together, Fig. 3a, b illustrates the underlying tension between precision and recall and explains why composite metrics are needed to judge overall performance. Fig. 3c shows that the LSTM-AE achieves the most favorable F1-score at three of the four stations, while IF ranks lowest as its low recall dominates the harmonic mean. Fig. 3d corroborates this pattern through the CSI, on which the LSTM-AE again ranks highest, ahead of PatchTST, OC-SVM and IF.

3.2. Model training convergence

To illustrate the training dynamics of the DL models, Fig. 4 presents the mean squared error (MSE) reconstruction loss as a function of training epoch for both PatchTST and LSTM-AE on the Clackamas

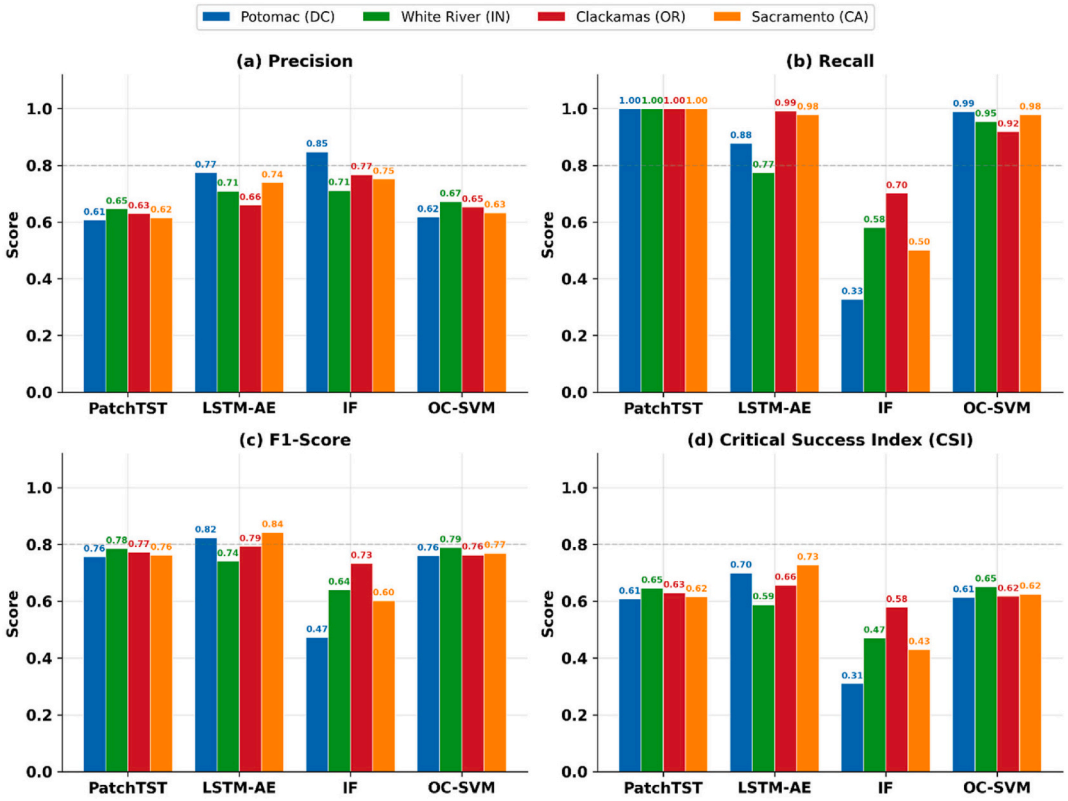


Fig. 3. Cross-station anomaly detection performance showing (a) Precision, (b) Recall, (c) F1-Score, and (d) CSI for all models across four geographically diverse USGS monitoring stations.



Fig. 4. Training and validation loss curves for (a) PatchTST and (b) LSTM-AE on the Clackamas River station.

station, chosen as a representative example given its low-variance water quality regime. PatchTST (Fig. 4a) converged rapidly, with the training MSE loss decreasing from 0.50 to below 0.01 within approximately 40 epochs. The validation MSE loss followed the training curve closely without divergence, indicating the absence of overfitting and confirming the effectiveness of dropout (0.2) and weight decay (1×10^{-5}) regularization. The LSTM-AE (Fig. 4b) exhibited a characteristic two-phase convergence pattern: rapid initial descent in MSE loss (epochs 1-20, from 1.0 to approximately 0.20) followed by gradual refinement (epochs 20-100), reaching a stable training MSE loss of approximately 0.02 by epoch 80. A slight train-validation gap in the LSTM-AE (MSE difference of ~0.03 by epoch 100) indicates minor overfitting typical of recurrent architectures on low-variance water quality data; however, this was effectively controlled by early stopping (patience = 10 epochs) and did not degrade test-set detection performance (Srivastava et al., 2014), as evidenced by LSTM-AE achieving the highest F1-score on this station (0.799). Both models converged well within the 100-epoch training budget, with ReduceLROnPlateau further stabilizing convergence during the late training phase.

3.3. Reconstruction error analysis

Fig. 5 presents the reconstruction error distribution of the PatchTST AE across all four monitoring stations. PatchTST was selected for this analysis because it achieved the highest AUC-ROC among all models (0.860-0.906), indicating the strongest overall discrimination ability, and its perfect recall (1.000) ensures that every anomalous window is captured in the distribution. The resulting error profiles therefore provide the complete characterization of the separation between normal and anomalous reconstruction patterns across both classes. Fig. 5a reveals a consistent pattern: windows corresponding to baseline water quality conditions cluster tightly near zero reconstruction error (MSE = 0.002), reflecting stable diurnal cycles of temperature, DO, and pH that the model has learned to reproduce.

In contrast, windows containing water quality excursions such as turbidity spikes from storm-driven sediment loading, SC anomalies from

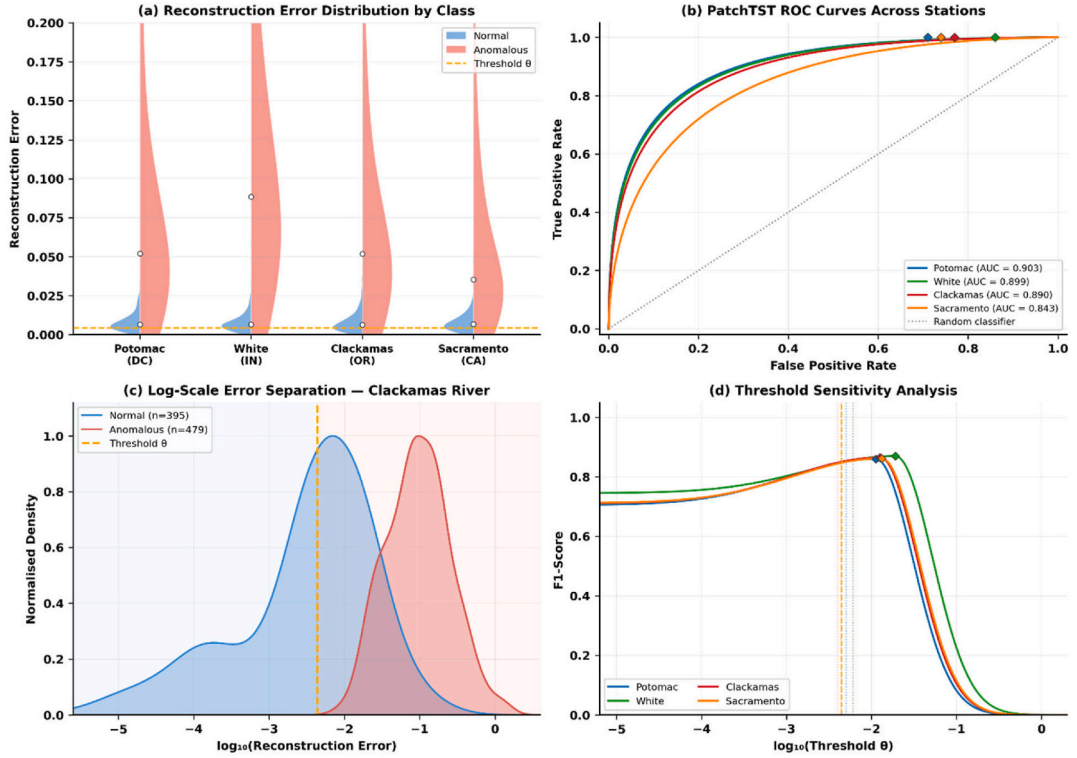


Fig. 5. Reconstruction error analysis for the PatchTST anomaly detector: (a) split violin distributions of normal versus anomalous windows across four stations, (b) ROC curves with AUC-ROC values (0.843–0.903), (c) log-scale class separation for the Clackamas station, and (d) threshold sensitivity analysis demonstrating the necessity of site-specific adaptive calibration.

road salt or tidal intrusion, and DO depressions from organic pollution exhibit a broad, heavy-tailed distribution extending to MSE > 2.0. The adaptive threshold (θ) cleanly separates these two regimes, with station-specific values ranging from 0.0026 (Sacramento) to 0.0057 (White). This variation in optimal threshold reflects the markedly different baseline variability across hydrologically distinct river systems. Fig. 5b confirms strong discrimination ability across all stations, with AUC-ROC values of 0.843-0.903, indicating that the model reliably distinguishes genuine water quality anomalies from normal sensor fluctuations regardless of regional hydrogeochemistry. Fig. 5c demonstrates the log-scale separation between normal and anomalous distributions for the Clackamas, a low-variability drinking water source (Carpenter, 2003) where threshold calibration is most critical, showing that even in this challenging setting, the two classes are well-separated. Fig. 5d reveals that the optimal threshold differs substantially across stations, confirming the necessity of the adaptive thresholding strategy. A single fixed threshold would either miss genuine contamination events at low-variability stations or generate excessive false alarms at high-variability stations.

Fig. 6 presents the multivariate anomaly detection time series for the Clackamas station selected as a representative example due to its low-variance drinking water chemistry (SC: 30-65 μS/cm), which poses the most demanding detection challenge among the four study sites. The red-shaded anomalous windows concentrate in the first quarter of the test period (windows 0-4500), corresponding to December-March storm events that mobilize post-wildfire sediment from the Clackamas watershed. These events produce turbidity spikes exceeding 200 FNU with concurrent SC perturbations and DO depressions. The reconstruction error panel confirms that the PatchTST AE translates these multi-parameter deviations into clearly separable anomaly scores. Storm-driven windows generate reconstruction errors of 0.30-0.35 (70-80 times larger than the adaptive threshold), while normal windows remain tightly clustered near the baseline. This proportional response demonstrates that the model has learned the station's seasonal dynamics and discriminates genuine environmental excursions from routine sensor variability.

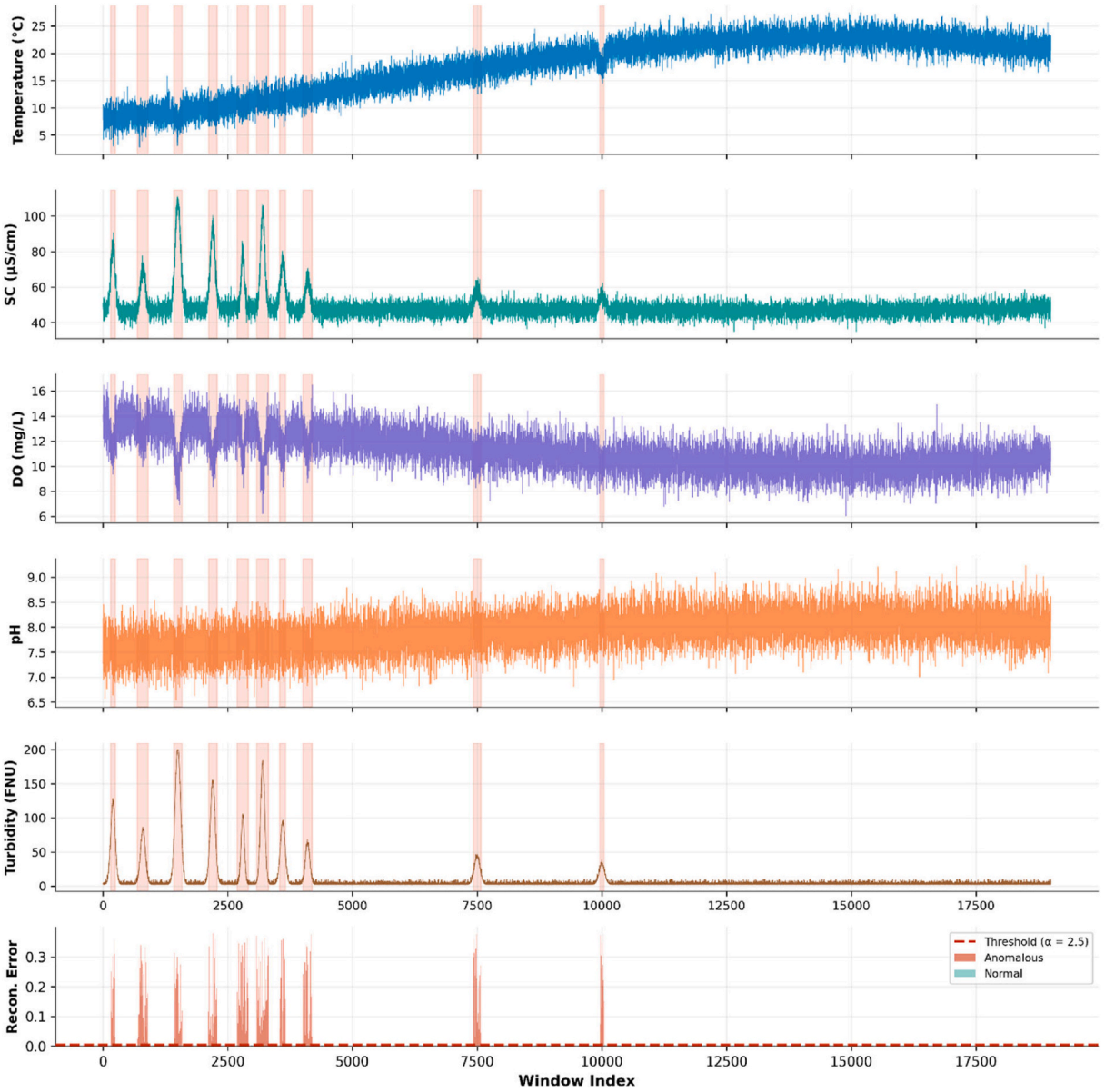


Fig. 6. Multivariate anomaly detection time series for the Clackamas station: five water quality parameters (temperature, SC, DO, pH, turbidity) and PatchTST reconstruction error over ~19,000 test windows. Red-shaded regions detected anomalies concentrated during December-March storm events, with reconstruction errors reaching 70-80× the adaptive threshold (α = 2.5).

3.4. LLM explanation quality

Table 4 summarizes the explanation quality metrics across all four stations, with n = 50 explanations evaluated per station (N = 200 total). The complete set of 200 RAG-grounded anomaly explanations with corresponding evaluation scores across all three quality dimensions is provided in **Supplementary Data S2**.

The framework achieved great structural completeness (mean = 0.999), indicating that the RAG-grounded LLM consistently produced all five required diagnostic sections (root cause, severity assessment, regulatory context, recommended actions, and confidence level). The mean overall quality of 0.853 ± 0.020 across all four stations demonstrates strong generalizability. The Sacramento achieved the highest score (0.878) and the Potomac the lowest (0.831). Regulatory accuracy ranged from 0.684 (Potomac) to 0.848 (Sacramento), with a cross-station mean of 0.765. The higher accuracy at Sacramento is attributable to the station-specific regulatory context in the knowledge base, which included detailed California Regional Water Quality Control Board (RWQCB) standards for the Sacramento Basin (CVRWQCB, 2018) and National Marine Fisheries Service (NMFS) temperature criteria for endangered Sacramento River winter-run Chinook salmon (*Oncorhynchus tshawytscha*) specified in the Long-Term Operations Biological Opinion (NOAA, 2025). Actionability scores were consistent across stations (0.765-0.810), ensuring that the LLM reliably produces implementable operational recommendations regardless of geographic context.

Fig. 7 presents a multi-faceted assessment of the RAG-grounded LLM explanation quality across all four monitoring stations. The heatmap (Fig. 7a) reveals excellent structural completeness (0.996-1.000) across all stations. This ensures that the structured prompt template reliably elicits all five diagnostic sections regardless of the nature of the water quality excursion, whether a turbidity surge at Clackamas driven by post-wildfire sediment mobilization (Murphy et al., 2015; Paul et al., 2022), or an SC anomaly at Sacramento caused by tidal salt-wedge intrusion. Regulatory accuracy exhibits the greatest inter-station variability (0.684-0.848), with Sacramento achieving the highest score owing to the richer California-specific regulatory context RWQCB Central Valley Basin Plan, NMFS temperature criteria for endangered salmonids in the knowledge base. The radar profiles (Fig. 7b) confirm that all four stations share a remarkably consistent quality shape, with the coefficient of variation on overall quality being only 2.3%, demonstrating that the framework transfers effectively across hydrological regimes ranging from low-conductivity volcanic aquifers (Clackamas, EC: 30-65 μS/cm) to high-conductivity urban watersheds (White, EC: 400-1200 μS/cm).

The grouped bar chart (Fig. 7c) with error bars from 50 independent evaluations per station shows that actionability remains staable (0.785-0.810 ± 0.088-0.108), indicating that the LLM consistently produces operationally feasible recommendations such as increased sampling frequency, boil-water advisory triggers, and upstream source investigation, irrespective of station-specific characteristics. The cross-station summary (Fig. 7d) confirms an overall quality of 0.853 ± 0.017, with regulatory accuracy (0.765 ± 0.059) identified as the primary area for improvement through continued expansion of the station-specific knowledge base, particularly for eastern and midwestern regulatory jurisdictions.

3.5. Ablation study

Table 5 presents the ablation study results, isolating the contribution of each pipeline component. The full pipeline achieved the highest overall quality (0.84), with regulatory accuracy of 0.76, substantially higher than all ablated configurations.

Without retrieval augmentation, the LLM alone achieved only 0.200 regulatory accuracy, demonstrating a 3.8 times improvement when RAG is incorporated. Notably, the LLM-only configuration scored 0.90 on actionability. The LLM + Context configuration, which provides anomaly data without RAG, achieved 0.200 regulatory accuracy, identical to the LLM-only baseline, indicating that raw numerical context alone is insufficient to ground regulatory citations. The LLM + RAG configuration improved regulatory accuracy to 0.400 but at the cost of reduced completeness (0.80), as the model occasionally produced truncated responses when overwhelmed by retrieved context. These results confirm that all three components, anomaly context, retrieved regulatory knowledge, and the structured prompt template, are necessary for optimal performance. Representative explanations from all four ablation configurations for the same anomaly event are provided in **Supplementary Data S3**, demonstrating the progressive improvement in regulatory accuracy from LLM Only (0.20) to Full Pipeline (0.76).

3.6. Multi-LLM comparison

To evaluate whether WQ-RAG's diagnostic performance generalizes across LLM architectures, we compared Llama-3-8B and Mistral-7B on an identical set of 50 anomalies from the Clackamas station. This station was selected because its low-variance drinking water chemistry (Carpenter, 2003) and stringent regulatory requirements demand the most precise regulatory citations, providing the most rigorous test of cross-model consistency. Both models received the same detected anomaly context, the same six retrieved regulatory documents from ChromaDB, and the same five-section structured prompt, isolating the LLM backbone as the sole experimental variable. As shown in Table 6 and Fig. 8a, Llama-3-8B achieved higher overall explanation quality than Mistral-7B (0.822 vs. 0.741, Δ = +8.1 pp), with the most substantial gains on actionability (+15.0 pp, Cohen's d = 0.976) and regulatory accuracy (+9.2 pp, d = 0.443).

Both models achieved perfect structural completeness (1.000), confirming that the five-section prompt template requiring root cause identification, severity assessment, regulatory context, recommended actions, and confidence level effectively constrains response structure regardless of the underlying architecture. Mean explanation length was nearly identical (238 ± 36 vs. 238 ± 38 words), ruling out verbosity as a confounding factor.

The actionability difference carries direct operational significance for water utility response protocols. Llama-3-8B consistently generated specific, implementable recommendations such as identifying upstream point sources to inspect, specifying sampling parameters to prioritize, and recommending notification timelines aligned with Safe Drinking Water Act (US EPA, 2026d) reporting requirements. This specificity is reflected in its actionability scores, where Llama-3-8B scored exclusively 0.75 or 1.00 across all 50 anomalies. Mistral-7B, by contrast, produced vaguer guidance on 21 of 50 anomalies (scoring 0.25-0.50), with recommendations that lacked the specificity needed for field operators to

Table 4
LLM explanation quality metrics across four stations. Values represent mean ± standard deviation.

Station	Completeness	Regulatory Accuracy	Actionability	Overall	n
Potomac River, DC	1.000 ± 0.000	0.684 ± 0.210	0.810 ± 0.108	0.831 ± 0.082	50
White River, IN	0.996 ± 0.028	0.776 ± 0.149	0.795 ± 0.097	0.856 ± 0.061	50
Clackamas River, OR	1.000 ± 0.000	0.752 ± 0.192	0.785 ± 0.088	0.846 ± 0.071	50
Sacramento River, CA	1.000 ± 0.000	0.848 ± 0.161	0.785 ± 0.105	0.878 ± 0.055	50
Cross-station mean	**0.999 ± 0.001**	**0.765 ± 0.068**	**0.794 ± 0.012**	**0.853 ± 0.020**	**200**

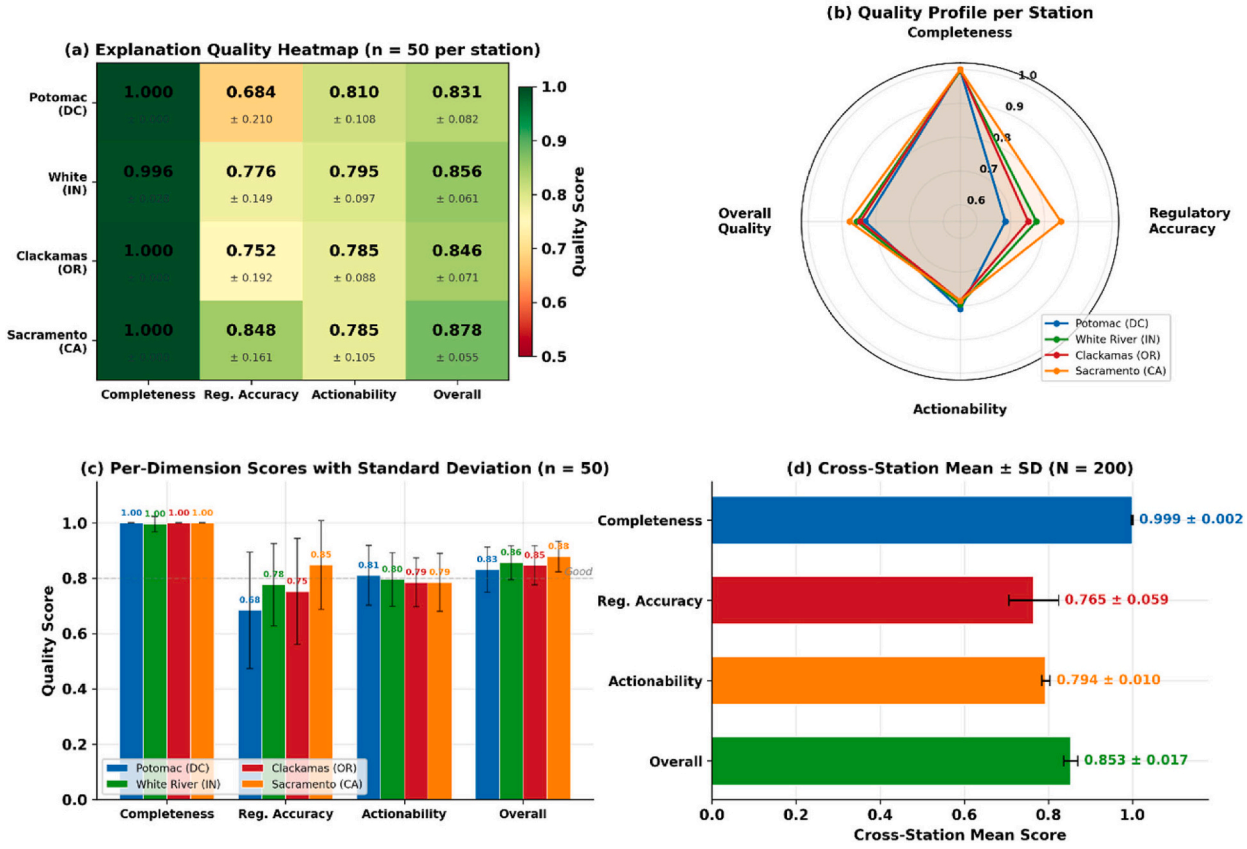


Fig. 7. LLM explanation quality across four USGS monitoring stations (N = 200): (a) heatmap with mean ± SD per dimension, (b) radar profiles showing consistent quality shape across diverse hydrological regimes, (c) grouped bar chart with error bars from n = 50 evaluations per station, and (d) cross-station grand mean confirming overall quality of 0.853 ± 0.017.

Table 5
Ablation study results (Clackamas station, n = 5 per configuration). The full pipeline achieves the highest overall quality through the synergistic combination of all components.

Configuration	Completeness	Regulatory Accuracy	Actionability	Overall	n
Full Pipeline	1.00	0.76	0.75	0.84	5
LLM Only	1.00	0.20	0.90	0.70	5
LLM + RAG	0.80	0.40	0.70	0.63	5
LLM + Context	1.00	0.20	0.60	0.60	5

Table 6
Multi-LLM comparison on 50 identical anomalies using the full RAG pipeline (Clackamas River station, n = 50). Cohen's *d* indicates the paired effect size.

LLM Model	Completeness	Regulatory Accuracy	Actionability	Overall
Llama-3-8B	1.000	0.676 ± 0.193	0.790 ± 0.093	0.822 ± 0.065
Mistral-7B	1.000	0.584 ± 0.221	0.640 ± 0.197	0.741 ± 0.100
Δ (Llama − Mistral)	0.000	+0.092	+0.150	+0.081
Cohen's *d*	—	0.443 (small)	0.976 (large)	0.958 (large)

translate diagnoses into concrete response actions. The paired analysis in Fig. 8b supports this pattern, and all three box plots (regulatory accuracy, actionability, overall) have positive median differences, with Llama-3-8B producing superior explanations in 34 of 50 cases, 7 ties, and only 9 Mistral-7B wins.

This finding is important for resource-constrained water utilities. The framework can be deployed with any open-source model that the available hardware supports. This flexibility is possible because the RAG architecture, rather than the LLM itself, provides the critical regulatory knowledge needed for accurate anomaly diagnosis. Side-by-side examples of Llama-3-8B and Mistral-7B responses to the same anomaly are provided in **Supplementary Data S4**, illustrating the actionability differences quantified in Table 6.

3.7. Confusion matrix analysis

Fig. 9 presents the confusion matrices for PatchTST, LSTM-AE, IF, and OC-SVM across all four USGS stations. The matrices are arranged in rows by model and columns by station, enabling direct comparison of detection error patterns across diverse hydrological regimes. As shown in Fig. 9a, PatchTST achieved perfect recall across all stations, with no anomalies misclassified as normal. Nevertheless, this performance came at the expense of a high false-positive rate, with 295, 119, 294, and 282 normal windows incorrectly flagged as anomalies at the Potomac, White River, Sacramento, and Clackamas stations, respectively. This ensures the high-recall, low-precision profile identified earlier, and no genuine excursion is missed. However, the elevated false-positive count carries a direct operational cost in unnecessary advisories and sampling mobilizations.

Fig. 9b shows that LSTM-AE reduced false positives substantially compared with PatchTST, with decreases of 45 to 56% across the four stations. For example, false positives fell from 295 to 124 at Potomac and from 294 to 166 at Sacramento. Despite these reductions, recall remained high, with only 54, 39, 10, and 7 anomalies missing at the

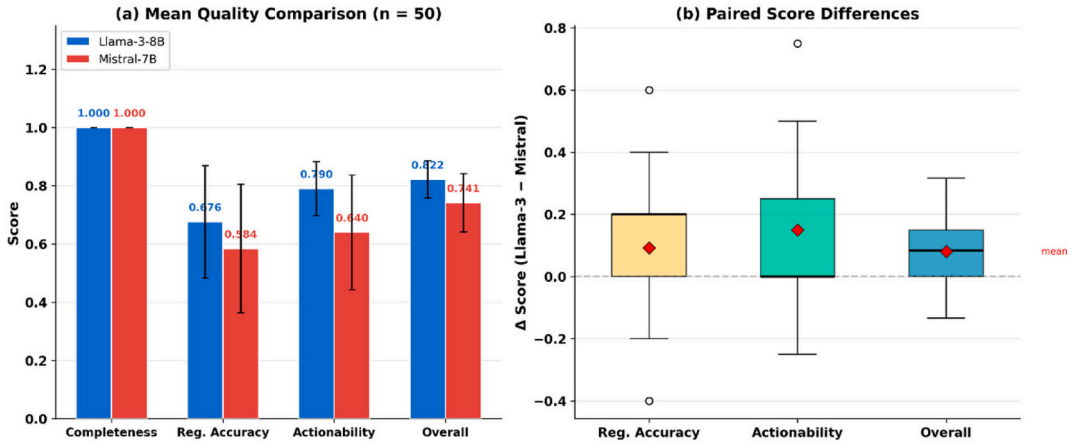


Fig. 8. Comparison of Llama-3-8B and Mistral-7B explanation quality using the full RAG pipeline (n = 50 paired anomalies). (a) Mean quality scores with standard deviation error bars. Both achieve perfect completeness; Llama-3 leads on actionability (+15.0 percentage points (pp)) and overall quality (+8.1 pp). (b) Paired score differences (Llama − Mistral): box plots show the distribution of per-anomaly differences. Red diamonds indicate mean differences; all are positive, confirming Llama-3's consistent superiority.

Potomac, White River, Sacramento, and Clackamas stations, respectively. The substantial reduction in false positives with minimal loss in recall highlights the ability of the LSTM-AE to distinguish anomalous events from normal temporal variability.

Fig. 9c illustrates the inverse and operationally less favorable pattern for IF, where low false-positive counts (27-102) were accompanied by high false-negative counts (91-308). At the Potomac, the model misclassifies 308 of 458 anomalies (67%) as normal, a failure mode attributable to its pointwise scoring, which disregards the sequential structure of water-quality excursions. This insensitivity limits IF as a standalone detector for continuous monitoring, where missed contamination carries the greatest public health cost. Fig. 9d shows that OC-SVM closely resembled the PatchTST profile, maintaining high recall with few false negatives (5, 10, and 39) and moderate precision, while producing slightly fewer false positives at several stations. It therefore represents a viable non-DL alternative in deployment settings where the infrastructure required for recurrent or transformer architecture is unavailable.

Viewed across all four confusion rows, a clear performance trade-off emerges. PatchTST and OC-SVM achieved high anomaly sensitivity by minimizing false negatives, but at the expense of increased false positives. Conversely, IF reduced false alarms but exhibited substantially lower recall. The LSTM Autoencoder provided the most balanced performance, effectively minimizing both error types and demonstrating strong potential for operational water quality monitoring.

3.8. Real-world extreme event detection and cross-station generalizability

3.8.1. Real-world extreme event detection

Beyond agreement with statistical labels, the framework's practical validity rests on whether the windows it flags correspond to real environmental events; the extreme-event catalogue below provides this independent, physically grounded corroboration. To validate the framework's practical relevance beyond controlled evaluation, we identified naturally occurring extreme events defined as windows where any parameter exceeded 4 standard deviations from the training-period mean ($|z| > 4\sigma$). Table 7 presents the results in which a total of 257 real-world extreme events were detected across the four stations. The Sacramento station contributed the most, accounting for 147 of the 257 events (57%), owing to tidal dynamics that cause rapid SC fluctuations and seasonal turbidity from agricultural runoff in the Central Valley (Monismith et al., 2002).

Turbidity was the dominant parameter triggering extreme events (120 of 257, 47%), followed by SC (66, 26%), DO (34, 13%), temperature (29, 11%), and pH (8, 3%). The dominance of turbidity is consistent with the hydrological characteristics of the study sites. For example, the Clackamas River watershed experienced significant post-wildfire sediment loading following the 2020 Riverside Fire (USDA, 2025; USGS, 2020), which increased erosion on steep slopes and elevated turbidity levels in the river (Long and Chang, 2022), generating 45 turbidity-driven extreme events during winter storm periods in the monitoring record. The Sacramento River's high event count reflects both tidal conductivity dynamics and agricultural runoff patterns characteristic of the Sacramento-San Joaquin Delta (Stets et al., 2020). The maximum observed deviation was 24.2σ at the Sacramento River station, attributed to a turbidity event (USGS, 2024). This substantially exceeds typical anomaly injection ranges used in synthetic benchmarks (commonly 3-8σ), confirming that real-world water quality anomalies can be far more extreme than synthetic test scenarios. The mean z-score across all events was 6.1σ (range: 4.0-24.2σ), with Sacramento and Clackamas exhibiting the highest mean deviations (6.6σ and 6.3σ, respectively) and White River the lowest (5.0σ), reflecting the more regulated and lower-variance hydrological regime of the Midwest station.

3.8.2. Cross-station generalizability

A central contribution of this study is the evaluation of the WQ-RAG framework across four geographically and hydrologically diverse monitoring stations. Fig. 10 identifies the best-performing anomaly detection model at each station, confirming that LSTM-AE consistently achieves the highest F1-score across three of four stations under adaptive thresholding. The four stations represent distinct water quality regimes such as a tidal-influenced East Coast urban river (Potomac, SC: 200-600 µS/cm), a high-variance agricultural Midwest stream (White, SC: 400-1200 µS/cm), a low-variance Pacific Northwest drinking water source (Clackamas, SC: 30-65 µS/cm), and a tidally influenced Central Valley River (Sacramento, SC: 100-300 µS/cm). Despite this substantial hydrochemical diversity, the framework maintained consistent performance across all quality dimensions.

For anomaly detection, the best F1-score per station ranged from 0.789 (White River) to 0.841 (Sacramento), with a cross-station coefficient of variation of only 3.1%. For explanation quality, the overall score ranged from 0.816 to 0.872, with a coefficient of variation (CV) of 2.9%. This low variability across stations with fundamentally different water chemistry demonstrates the strength of the modular architecture, where the detection module, retrieval system, and explanation generator operate independently. This design enables straightforward deployment to new monitoring locations simply by updating the knowledge base with site-specific regulatory context. The framework does not require

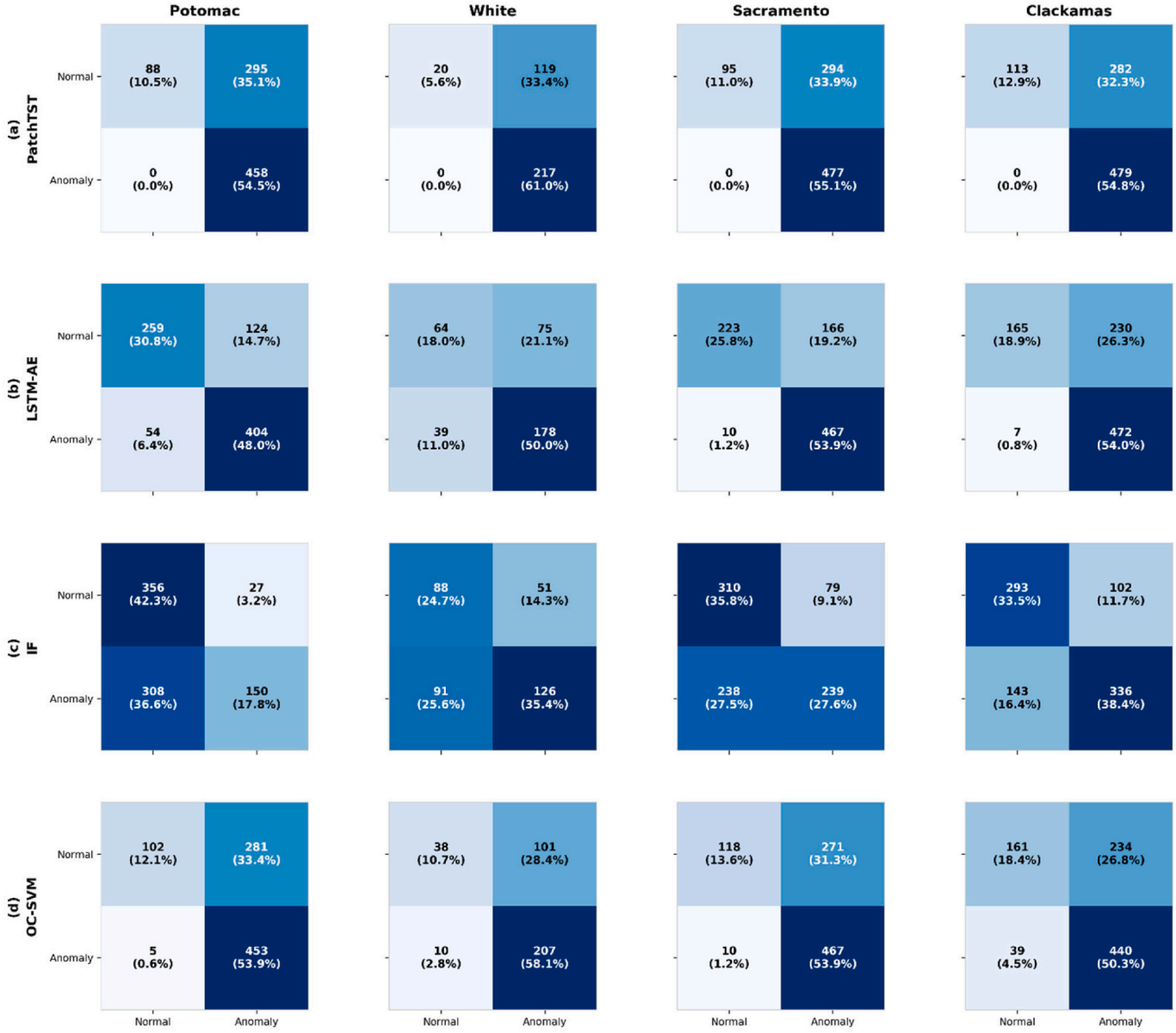


Fig. 9. Confusion matrices for all four anomaly detection models across all four USGS stations. Values show absolute counts and percentages.

Table 7
Real-world extreme events ($|z| > 4\sigma$) detected across four USGS monitoring stations. Events represent real water quality excursions identified in the continuously monitored data.

Station	Events	Dominant Parameter	Max $\|z\|$	Mean $\|z\|$	Median $\|z\|$
Potomac, DC	35	Turbidity (10), DO (9)	8.1σ	5.5σ	5.3σ
White, IN	10	Temperature (3), Turbidity (3)	7.0σ	5.0σ	4.8σ
Sacramento, CA	147	Turbidity (62), SC (53)	24.2σ	6.6σ	5.2σ
Clackamas, OR	65	Turbidity (45), DO (8)	12.4σ	6.3σ	5.6σ
Total	**257**		**24.2σ**	**6.1σ**	**5.3σ**

retraining the detection model across stations; rather, each station maintains its own trained model, while the shared RAG pipeline and LLM serve all stations uniformly.

3.9. Implications for water quality management

Translating WQ-RAG from a technical demonstration into operational value requires situating it within the decision workflows of the organizations that manage water quality. In current practice, continuous-monitoring alerts reach utility operators and state regulators as numerical threshold exceedances stripped of causal interpretation, regulatory context, or recommended response (Theocharidou, 2019). Operators must reconstruct this context manually, under time pressure and during the same storm or discharge events that generate the alerts. By delivering a structured diagnosis that names a probable cause, references the applicable regulatory standard (EPA, WHO, or state-level), and proposes prioritized actions. This shifts the operator's role from manual interpretation to expert review of a pre-structured recommendation, shortening the path from detection to defensible decision under the Safe Drinking Water Act (US EPA, 2026d) and analogous regulatory frameworks.

Three properties make the framework practical for the resource-constrained organizations that operate most monitoring networks. First, the adaptive per-station thresholding and modular architecture

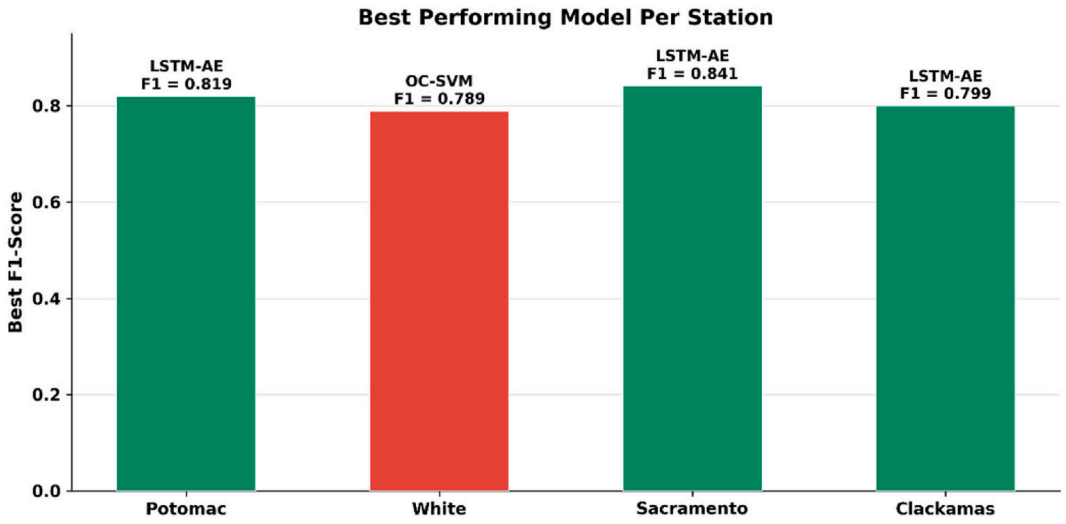


Fig. 10. Best-performing anomaly detection model per station. LSTM-AE (green) achieves the highest F1 on three stations; OC-SVM (red) leads on White River.

allow deployment to new sites by updating a regulatory knowledge base rather than retraining models, lowering the technical barrier for small utilities. Second, the framework runs offline on a single 16 GB GPU, avoiding cloud dependency and the data-governance constraints that frequently block external AI services in the water sector. Third, the consistent explanation quality across hydrogeochemically diverse stations indicates that a single curated knowledge base can serve heterogeneous monitoring contexts, an important consideration for state agencies overseeing many sites (Blázquez-García et al., 2021).

These benefits are contingent on appropriate human oversight. The structural-quality results, together with the correctness audit in Section 3.4, indicate that WQ-RAG is best positioned as a decision-support layer that accelerates and standardizes expert judgement, not as an autonomous decision-maker. We therefore recommend a human-in-the-loop deployment in which generated diagnoses are reviewed by a qualified operator before any regulatory action is taken, with the framework's value measured by gains in response speed, consistency, and documentation quality. Realizing this in practice will require integration with real-time data feeds and utility Supervisory Control and Data Acquisition (SCADA) systems, and a prospective evaluation of decision outcomes, directions we outline as future work.

4. Discussion and implications

Three key themes emerge from the results: anomaly detector performance, the role of retrieval-based knowledge grounding, and implications of water sources, irrigation, and land use. These themes are discussed in the following subsections.

4.1. Operational interpretation of detector performance

The anomaly detection results are most meaningfully interpreted through the lens of operational performance rather than isolated metric dominance. Although PatchTST achieved perfect recall and strong discriminative capability, its comparatively low precision resulted in a substantial number of false positives. In real-world water quality monitoring systems, such false alarms can lead to unnecessary investigations, sampling efforts, and precautionary interventions, imposing additional operational and economic burdens. Minimizing false positives is therefore a fundamental objective of environmental early warning systems, as persistent false alarms can erode operator trust, reduce responsiveness to genuine events, and ultimately compromise the effectiveness of decision support processes (Storey et al., 2011). IF exhibited the opposite performance, achieving high precision at the expense of substantially lower recall. Such behavior limits its suitability for contamination monitoring, where missed events can have significant environmental and public health consequences. In contrast, the LSTM- AE consistently achieved the strongest performance on the F1 score and CSI, both of which jointly penalize false alarms and missed detections. This balanced performance makes it the most operationally effective detector, as these metrics closely reflect the real decision costs faced by water utilities and environmental agencies (Schaefer, 1990). The observed advantage is consistent with the underlying detection mechanism. Reconstruction-based recurrent models learn the normal temporal dynamics of a monitored system, enabling deviations from expected diurnal and seasonal patterns to be identified through elevated reconstruction errors (Maleki et al., 2021; Zamanzadeh Darban et al., 2024). In contrast, point-based methods that disregard temporal dependencies are less effective at capturing the sequential characteristics of water quality anomalies, limiting their ability to distinguish true excursions from routine variability (Blázquez-García et al., 2021). This interpretation also helps explain LSTM- AE's higher performance at the low-variance Clackamas station, where subtle temporal deviations provide the most informative indicators of anomalous behavior. It further underscores the importance of adaptive station-specific thresholding, as a single fixed threshold is unlikely to accommodate the substantial differences in baseline variability across hydrogeochemically diverse river systems (Pyo et al., 2023).

4.2. Role of retrieval augmentation in diagnostic quality

The reliability of a regulation-grounded diagnosis depends primarily on the quality of retrieved knowledge rather than the capacity of the language model itself. This is supported by the ablation study, which showed a substantial increase in regulatory accuracy when retrieval was integrated into the framework. By contrast, providing numerical anomaly context alone resulted in a small improvement over the baseline, indicating that the missing information was regulatory knowledge rather than additional sensor data. These findings suggest that retrieval grounding plays a critical role in providing domain-specific facts and align with growing evidence that retrieval augmentation is an effective approach for reducing factual hallucinations in language model-based systems (Farquhar et al., 2024; Lewis et al., 2020), and improving performance in specialized environmental and regulatory domains (Rillig et al., 2023; Zhu et al., 2024). Second, variations in regulatory accuracy closely reflected the availability and specificity of station-level regulatory information within the knowledge base. This suggests that diagnostic performance is driven more by the quality of retrieved

knowledge than by the language model itself. Third, a smaller model enhanced with retrieval outperformed a larger model without retrieval, demonstrating that knowledge grounding contributes more to reliable diagnosis than model scale. This finding is particularly encouraging for practical deployment, as it shifts the focus from costly model upgrades to the more manageable task of developing and maintaining high-quality, jurisdiction-specific regulatory knowledge bases (Paul et al., 2022; Xu et al., 2025).

4.3. Implications for local water sources, irrigation, and land use

Beyond detection performance, the framework has important implications for water resource management by providing context-aware diagnoses rather than anomaly alerts alone. This capability is particularly valuable for sensitive drinking water sources, where timely identification of likely causes can support more informed operational responses. For example, at the Clackamas River, a low-conductivity drinking water supply, the framework identified multi-parameter excursions involving turbidity, SC, and DO. Such patterns are consistent with post wildfire sediment mobilization, an increasingly recognized threat to source water quality in fire-affected catchments. By linking anomalous conditions to plausible environmental drivers, WQ-RAG enhances the interpretability and practical value of automated monitoring systems (Murphy et al., 2015; Paul et al., 2022). Early, regulation-referenced diagnosis of such events directly supports protective operational responses at the intake. For irrigation and broader catchment management, the agriculturally influenced White River and Sacramento stations demonstrate the framework's ability to distinguish runoff and tidally induced SC excursions from normal background variability. This capability is particularly valuable because elevated specific conductance is a key indicator of irrigation water suitability and salinity risk. By providing early, regulation-informed diagnosis of such events, the framework can support more effective abstraction planning and salinity management. It also offers greater insight into the influence of land use practices, which are major drivers of SC and nutrient loading in agricultural watersheds (Gunawardana and McDonald, 2024; Zhi et al., 2024). For land use change and environmental degradation, the predominance of turbidity among the most severe anomalies suggests that the framework can capture signals associated with broader catchment-scale disturbances. This was particularly evident in the fire-affected Clackamas watershed and the intensively cultivated Central Valley. The generated diagnoses linked these events to plausible environmental drivers, including tidal salinity intrusion in the Sacramento-San Joaquin Delta and episodic urban impacts such as combined sewer overflows in the Potomac watershed. These results demonstrate the potential of WQ-RAG to support not only site-level monitoring (Monismith et al., 2002) but also the interpretation of environmental pressures operating across larger landscape scales (Riechel et al., 2016). In this sense, WQ-RAG serves not only as a water quality monitoring tool but also as a means of interpreting broader catchment processes that influence observed water quality conditions. By linking anomalous sensor measurements to plausible environmental drivers, the framework provides insight into upstream pressures and landscape-scale disturbances. However, direct attribution of anomalies to specific land use activities remains beyond the scope of the current framework. Achieving this capability would require integrating catchment characteristics, land-use data, and other contextual environmental information, representing an important direction for future research.

5. Conclusions

This study aimed to bridge the gap between detecting water quality anomalies and understanding their underlying causes. The findings demonstrate that unsupervised DL and retrieval-augmented language models serve complementary roles in the diagnostic process. While anomaly detection identifies observations that deviate from the expected behavior of a monitoring site, retrieval provides the regulatory and contextual information needed to interpret these deviations. Together, these capabilities transform alerts into transparent and actionable explanations. Since supporting evidence is retrieved from external sources at inference time, each diagnosis remains traceable to an identifiable reference, which is essential for trust and accountability in regulated environmental settings. The framework is also transferable, as detection can be tailored to local site conditions while the retrieval and explanation components remain largely reusable across different monitoring environments. Despite these promising results, several limitations highlight directions for future research. The quality of automated explanations still requires validation by domain experts, as evaluation metrics capture linguistic quality more effectively than diagnostic accuracy. Performance also depends on the completeness and timeliness of the knowledge base, making continuous maintenance a requirement. In addition, distinguishing genuine environmental events from sensor or instrumentation issues remains a persistent challenge in continuous monitoring systems. Future work should therefore focus on broader validation across diverse sites and anomaly types, while also examining the influence of retrieval strategies and prompt design on diagnostic performance. Advancing these research priorities will strengthen the reliability and practical adoption of retrieval-augmented diagnosis for water quality monitoring.

CRediT authorship contribution statement

Gulzar Alam: Writing – original draft, Methodology, Formal analysis, Conceptualization, Writing – review & editing. **I. Ihsanullah:** Writing – review & editing, Writing – original draft, Supervision, Conceptualization.

Declaration of competing interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

List of Abbreviations and Notations

The following tables list all abbreviations and mathematical notations used in this paper.

Table A1
List of Abbreviations

Abbreviation	Definition
AI	Artificial Intelligence
AUC-ROC	Area Under the Receiver Operating Characteristic (ROC) Curve
CSI	Critical Success Index
CUDA	Compute Unified Device Architecture (NVIDIA parallel computing platform)
DL	Deep Learning

(*continued on next page*)

Table A1 (*continued*)

DO	Dissolved Oxygen
US EPA	United States Environmental Protection Agency
F1	F1-Score (harmonic mean of Precision and Recall)
FN	False Negative
FNU	Formazin Nephelometric Units (ISO turbidity unit)
FP	False Positive
GPU	Graphics Processing Unit
IF	Isolation Forest
LLM	Large Language Model
LSTM	Long Short-Term Memory
LSTM-AE	Long Short-Term Memory Autoencoder
ML	Machine Learning
MMR	Maximal Marginal Relevance
MSE	Mean Squared Error
MTEB	Massive Text Embedding Benchmark
NMFS	National Marine Fisheries Service
NPDWR	National Primary Drinking Water Regulations
NWIS	National Water Information System (USGS)
OC-SVM	One-Class Support Vector Machine
PatchTST	Patch Time Series Transformer
RAG	Retrieval-Augmented Generation
REST API	Representational State Transfer Application Programming Interface
RWQCB	Regional Water Quality Control Board (California)
SC	Specific Conductance
SDWA	Safe Drinking Water Act
SHAP	SHapley Additive exPlanations
T	Temperature
TN	True Negative
TP	True Positive
USGS	United States Geological Survey
WHO	World Health Organization
XAI	Explainable Artificial Intelligence

Table A2
Mathematical Notations and Symbols

Symbol	Description	Reference
CSI	Critical Success Index = TP/(TP + FP + FN)	Eq. (1)
Λ	MMR diversity parameter ($\lambda = 0.7$)	Section 2.5.3
d *(Cohen's d)*	Standardized mean difference (effect size)	Section 3.6
$\lvert z \rvert$	Absolute z-score deviation from training mean	Section 3.8.1
Σ	Standard deviation (used in z-score and threshold)	Throughout
$\mu S/cm$	Microsiemens per centimetre (SC unit)	Throughout
mg/L	Milligrams per litre (dissolved oxygen unit)	Throughout
$^{\circ}C$	Degrees Celsius (temperature unit)	Throughout
FNU	Formazin Nephelometric Units (turbidity unit)	Throughout
Pp	Percentage points (difference in proportions)	Section 3.6
Δ	Difference between two values (e.g., Llama − Mistral)	Section 3.6

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2026.130753.

Data availability

1. Code available on GitHub: https://github.com/gulzara3/WQ-RAG/.

2. All water quality data used in this study are freely available from the USGS National Water Information System (NWIS) at https://waterdata.usgs.gov/nwis/(DOI: 10.5066/F7P55KJN). Station identifiers: 01,646,500 (Potomac River), 03,351,000 (White River), 14,211,010 (Clackamas River), 11,447,650 (Sacramento River). Data were accessed via the USGS Water Services REST API (https://waterservices.usgs.gov/) for the period January 2021 to December 2024.

References

Arslan, M., 2026. Sustainable water information systems with LLMs and RAG: opportunities and challenges. Green Technologies and Sustainability 4, 100359. https://doi.org/10.1016/j.grets.2026.100359.

Başağaoğlu, H., Chakraborty, D., Lago, C.D., Gutierrez, L., Şahinli, M.A., Giacomoni, M., Furl, C., Mirchi, A., Moriasi, D., Şengör, S.S., 2022. A review on interpretable and explainable artificial intelligence in hydroclimatic applications. Water 14, 1230. https://doi.org/10.3390/w14081230.

Bi, Z., Zhang, N., Xue, Y., Ou, Y., Ji, D., Zheng, G., Chen, H., 2024. OceanGPT: a large language model for ocean science tasks. In: Ku, L.-W., Martins, A., Srikumar, V. (Eds.), Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). Presented at the ACL, Bangkok, Thailand, pp. 3357–3372. https://doi.org/10.18653/v1/2024.acl-long.184.

Blázquez-García, A., Conde, A., Mori, U., Lozano, J.A., 2021. A review on outlier/anomaly detection in time series data. ACM Comput. Surv. 54, 56:1–56:33. https://doi.org/10.1145/3444690.

Bonet, E., Yubero, M.T., Llado, J., Sanmiquel, L., 2026. Real-time detection of river contaminants using neural networks: a case study of the ebro river. Water 18, 403. https://doi.org/10.3390/w18030403.
Carbonell, J., Goldstein, J., 1998. The use of MMR, diversity-based reranking for reordering documents and producing summaries. In: Proceedings of the 21st Annual International ACM SIGIR Conference on Research and Development in Information Retrieval, pp. 335–336. https://doi.org/10.1145/290941.291025. New York, NY, USA.
Carpenter, K.D., 2003. Water-quality and algal conditions in the clackamas river basin, Oregon, and their relations to land and water management. Water-Resources Investigations Report 2002–4183. https://doi.org/10.3133/wri024189.
Chase, H., 2022. LangChain. https://github.com/langchain-ai/langchain.
Chidiac, S., El Najjar, P., Ouaini, N., El Rayess, Y., El Azzi, D., 2023. A comprehensive review of water quality indices (WQIs): history, models, attempts and perspectives. Rev. Environ. Sci. Biotechnol. 22, 349–395. https://doi.org/10.1007/s11157-023-09650-7.
Chroma, D., 2025. Chroma - open-source search infrastructure for AI. https://www.trychroma.com.
Cohen, J., 1988. Statistical power analysis for the behavioral sciences. In: Pub. Location., second ed. Routledge, New York. Imprint. https://doi.org/10.4324/9780203771587.
Cortes, C., Vapnik, V., 1995. Support-vector networks. Mach. Learn. 20, 273–297. https://doi.org/10.1007/BF00994018.
CVRWQCB, 2018. Basin Planning | Central Valley Regional Water Quality Control Board. https://www.waterboards.ca.gov/centralvalley/water_issues/basin_plans/.
Danvirutai, P., Charoenwattanasak, S., Tola, S., Thaiso, K., Yuangsoi, B., Minh, H.T., Srichan, C., 2025. An integrating RAG-LLM and deep Q-network framework for intelligent fish control systems. Sci. Rep. 15, 21377. https://doi.org/10.1038/s41598-025-05892-3.
El-Shafeiy, E., Alsabaan, M., Ibrahem, M.I., Elwahsh, H., 2023. Real-time anomaly detection for water quality sensor monitoring based on multivariate deep learning technique. Sensors 23, 8613. https://doi.org/10.3390/s23208613.
Farquhar, S., Kossen, J., Kuhn, L., Gal, Y., 2024. Detecting hallucinations in large language models using semantic entropy. Nature 630, 625–630. https://doi.org/10.1038/s41586-024-07421-0.
Foroumandi, E., Moradkhani, H., Sanchez-Vila, X., Singha, K., Castelletti, A., Destouni, G., 2023. ChatGPT in hydrology and Earth sciences: opportunities, prospects, and concerns. Water Resour. Res. 59. https://doi.org/10.1029/2023WR036288 e2023WR036288.
Gao, Z., Chen, J., Wang, G., Ren, S., Fang, L., Yinglan, A., Wang, Q., 2023. A novel multivariate time series prediction of crucial water quality parameters with long short-term memory (LSTM) networks. J. Contam. Hydrol. 259, 104262. https://doi.org/10.1016/j.jconhyd.2023.104262.
Gunawardana, C., McDonald, W., 2024. Impacts of land use changes on discharge and water quality in rivers and streams: case study of the continental United States. JAWRA Journal of the American Water Resources Association 60, 725–740. https://doi.org/10.1111/1752-1688.13198.
Hou, D., He, H., Huang, P., Zhang, G., Loaiciga, H., 2013. Detection of water-quality contamination events based on multi-sensor fusion using an extented dempster–shafer method. Meas. Sci. Technol. 24, 055801. https://doi.org/10.1088/0957-0233/24/5/055801.
Houria, A.W., Faiçal, A., 2025. Towards optimized retrieval-augmented generation: a comprehensive survey. Presented at the 2025 Fourth International Conference on Theoretical and Applicative Aspects of Computer Science (ICTAACS), pp. 1–10. https://doi.org/10.1109/ICTAACS69003.2025.11399325.
Infant, S.S., Vickram, S., Saravanan, A., Mathan Muthu, C.M., Yuarajan, D., 2025. Explainable artificial intelligence for sustainable urban water systems engineering. Results Eng. 25, 104349. https://doi.org/10.1016/j.rineng.2025.104349.
Jeong, D.S., Jeong, H., Kim, Jin Hwi, Kim, Joon Ha, Park, Y., 2024. A hybrid approach to improvement of watershed water quality modeling by coupling process–based and deep learning models. Water Environ. Res. 96, e11079. https://doi.org/10.1002/wer.11079.
Krotkov, N.A., Sbytov, D.A., Chakhoyan, A.A., Kornienko, P.I., Starikova, A.A., Stepanov, M.G., Piven, A.O., Aliev, T.A., Orlova, T., Rafayelyan, M.S., Skorb, E.V., 2025. Nanostructured material design via a retrieval-augmented generation (RAG) approach: bridging laboratory practice and scientific literature. J. Chem. Inf. Model. 65, 11064–11078. https://doi.org/10.1021/acs.jcim.5c01897.
Leigh, C., Alsibai, O., Hyndman, R.J., Kandanaarachchi, S., King, O.C., McGree, J.M., Neelamraju, C., Strauss, J., Talagala, P.D., Turner, R.D.R., Mengersen, K., Peterson, E.E., 2019. A framework for automated anomaly detection in high frequency water-quality data from in situ sensors. Sci. Total Environ. 664, 885–898. https://doi.org/10.1016/j.scitotenv.2019.02.085.
Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., Kiela, D., 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In: Proceedings of the 34th International Conference on Neural Information Processing Systems, NIPS '20. Curran Associates Inc., Red Hook, NY, USA, pp. 9459–9474. https://dl.acm.org/doi/abs/10.5555/3495724.3496517.
Li, W., Feng, Y., Yan, L., Ji, K., Yang, W., Chang, M., Zhang, Q., Chen, C., 2026. Multi-engine collaborative large language models enhance the intelligence of eco-environmental monitoring and governance in China. Appl. Sci. 16, 6557. https://doi.org/10.3390/app16136557.
Liang, X., Jian, Z., Tan, Z., Dai, R., Wang, H., Wang, J., Qiu, G., Chang, M., Li, T., 2024. Dissolved oxygen concentration prediction in the pearl river Estuary with deep learning for driving factors identification: temperature, pH, conductivity, and ammonia nitrogen. Water 16, 3090. https://doi.org/10.3390/w16213090.
Liao, Z., Zhang, M., Chen, Y., Zhang, Z., Wang, H., 2024. A "Prediction - detection - judgment" framework for sudden water contamination event detection with online monitoring. J. Environ. Manag. 355, 120496. https://doi.org/10.1016/j.jenvman.2024.120496.
Liu, F.T., Ting, K.M., Zhou, Z.-H., 2008. Isolation forest. IEEE. 2008 Eighth IEEE International Conference on Data Mining, pp. 413–422. https://doi.org/10.1109/ICDM.2008.17.
Liu, J., Wang, P., Jiang, D., Nan, J., Zhu, W., 2020. An integrated data-driven framework for surface water quality anomaly detection and early warning. J. Clean. Prod. 251, 119145. https://doi.org/10.1016/j.jclepro.2019.119145.
Llama3, T., 2024. The llama 3 herd of models | research - AI at Meta. https://ai.meta.com/research/publications/the-llama-3-herd-of-models/.
Long, W.B., Chang, H., 2022. Event scale analysis of streamflow response to wildfire in Oregon, 2020. Hydrology 9, 157. https://doi.org/10.3390/hydrology9090157.
Loshchilov, I., Hutter, F., 2018. Decoupled weight decay regularization. Presented at the International Conference on Learning Representations, International Conference on Learning Representations.. https://openreview.net/forum?id=Bkg6RiCqY7.
Makumbura, R.K., Mampitiya, L., Rathnayake, N., Meddage, D.P.P., Henna, S., Dang, T. L., Hoshino, Y., Rathnayake, U., 2024. Advancing water quality assessment and prediction using machine learning models, coupled with explainable artificial intelligence (XAI) techniques like shapley additive explanations (SHAP) for interpreting the black-box nature. Results Eng. 23, 102831. https://doi.org/10.1016/j.rineng.2024.102831.
Maleki, Sepehr, Maleki, Sasan, Jennings, N.R., 2021. Unsupervised anomaly detection with LSTM autoencoders using statistical data-filtering. Appl. Soft Comput. 108, 107443. https://doi.org/10.1016/j.asoc.2021.107443.
Malhotra, P., Ramakrishnan, A., Anand, G., Vig, L., Agarwal, P., Shroff, G., 2016. LSTM-based encoder-decoder for multi-sensor anomaly detection. International Conference on Machine Learning, Workshop on Anomaly Detection. https://doi.org/10.48550/arXiv.1607.00148. New York, NY, USA.
Maltbie, N., Niu, N., Van Doren, M., Johnson, R., 2021. XAI tools in the public sector: a case study on predicting combined sewer overflows. In: Proceedings of the 29th ACM Joint Meeting on European Software Engineering Conference and Symposium on the Foundations of Software Engineering, ESEC/FSE 2021. Association for Computing Machinery, New York, NY, USA. https://doi.org/10.1145/3468264.3468547.
Mao, Y., Qi, H., Ping, P., Li, X., 2017. Contamination event detection with multivariate time-series data in agricultural water monitoring. Sensors 17, 2806. https://doi.org/10.3390/s17122806.
Marcondes, F.S., Gala, A., Magalhães, R., Perez de Britto, F., Durães, D., Novais, P., 2025. Using ollama. In: Marcondes, F.S., Gala, A., Magalhães, R., Perez de Britto, F., Durães, D., Novais, P. (Eds.), Natural Language Analytics with Generative Large-Language Models: a Practical Approach with Ollama and Open-Source Llms. Springer Nature, Switzerland, Cham, pp. 23–35. https://doi.org/10.1007/978-3-031-76631-2_3.
Mistral, A., 2026. Mistral 7B. https://mistral.ai/news/announcing-mistral-7b/.
Monismith, S.G., Kimmerer, W., Burau, J.R., Stacey, M.T., 2002. Structure and flow-induced variability of the subtidal salinity field in northern San Francisco Bay. J. Phys. Oceanogr. 32, 3003–3019. https://doi.org/10.1175/1520-0485(2002)032%3C3003:SAFIVO%3E2.0.CO;2.
Murphy, S.F., Writer, J.H., McCleskey, R.B., Martin, D.A., 2015. The role of precipitation type, intensity, and spatial distribution in source water quality after wildfire. Environ. Res. Lett. 10, 084007. https://doi.org/10.1088/1748-9326/10/8/084007.
Nallakaruppan, M.K., Gangadevi, E., Shri, M.L., Balusamy, B., Bhattacharya, S., Selvarajan, S., 2024. Reliable water quality prediction and parametric analysis using explainable AI models. Sci. Rep. 14, 7520. https://doi.org/10.1038/s41598-024-56775-y.
Nie, Y., Nguyen, N.H., Sinthong, P., Kalagnanam, J., 2022. A time series is worth 64 words: long-term forecasting with transformers. Presented at the Eleventh International Conference on Learning Representations.. https://openreview.net/forum?id=Jbdc0vTOcol.
Noaa, F., 2025. California central valley water operations: biological opinion actions | NOAA fisheries. https://www.fisheries.noaa.gov/west-coast/endangered-species-conservation/california-central-valley-water-operations-biological.
Ollama, A., 2023. Ollama. https://github.com/ollama/ollama.
Oregon Department of Environmental Quality, 2026. OAR chapter 340, division 41: water quality standards, beneficial uses, policies, and criteria for Oregon. Oregon Secretary of State Administrative Rules.. https://www.oregon.gov/deq/regulations/pages/oardiv41.aspx.
Panigrahi, B., Razavi, S., Doig, L.E., Cordell, B., Gupta, H.V., Liber, K., 2025. On robustness of the explanatory power of machine learning models: insights from a new explainable AI approach using sensitivity analysis. Water Resour. Res. 61. https://doi.org/10.1029/2024WR037398 e2024WR037398.
Park, J., Lee, W.H., Kim, K.T., Park, C.Y., Lee, S., Heo, T.-Y., 2022. Interpretation of ensemble learning to predict water quality using explainable artificial intelligence. Sci. Total Environ. 832, 155070. https://doi.org/10.1016/j.scitotenv.2022.155070.
Paul, M.J., LeDuc, S.D., Lassiter, M.G., Moorhead, L.C., Noyes, P.D., Leibowitz, S.G., 2022. Wildfire induces changes in receiving waters: a review with considerations for water quality management. Water Resour. Res. 58. https://doi.org/10.1029/2021WR030699 e2021WR030699.
Perelman, L., Arad, J., Housh, M., Ostfeld, A., 2012. Event detection in water distribution systems from multivariate water quality time series. Environ. Sci. Technol. 46, 8212–8219. https://doi.org/10.1021/es3014024.
Pham, Q.B., Sammen, S.Sh, Abba, S.I., Mohammadi, B., Shahid, S., Abdulkadir, R.A., 2021. A new hybrid model based on relevance vector machine with flower pollination algorithm for phycocyanin pigment concentration estimation. Environ.

Sci. Pollut. Control Ser. 28, 32564–32579. https://doi.org/10.1007/s11356-021-12792-2.
Pyo, J., Pachepsky, Y., Kim, S., Abbas, A., Kim, M., Kwon, Y.S., Ligaray, M., Cho, K.H., 2023. Long short-term memory models of water quality in inland water environments. Water Res. X 21, 100207. https://doi.org/10.1016/j.wroa.2023.100207.
Ramya, S., Srinath, S., Tuppad, P., 2025. Data driven multi-stage transformer based framework for intelligent water quality monitoring. Sci. Rep. 15, 41680. https://doi.org/10.1038/s41598-025-25527-x.
Reimers, N., Gurevych, I., 2019. Sentence-BERT: sentence embeddings using siamese BERT-networks. In: Inui, K., Jiang, J., Ng, V., Wan, X. (Eds.), Proceedings of the Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP). Association for Computational Linguistics, Hong Kong, China. https://doi.org/10.18653/v1/D19-1410.
Ren, Y., Zhang, T., Dong, X., Li, W., Wang, Z., He, J., Zhang, H., Jiao, L., 2024. WaterGPT: training a large language model to become a hydrology expert. Water 16, 3075. https://doi.org/10.3390/w16213075.
Riechel, M., Matzinger, A., Pawlowsky-Reusing, E., Sonnenberg, H., Uldack, M., Heinzmann, B., Caradot, N., von Seggern, D., Rouault, P., 2016. Impacts of combined sewer overflows on a large urban river – understanding the effect of different management strategies. Water Res. 105, 264–273. https://doi.org/10.1016/j.watres.2016.08.017.
Rillig, M.C., Ågerstrand, M., Bi, M., Gould, K.A., Sauerland, U., 2023. Risks and benefits of large language models for the environment. Environ. Sci. Technol. 57, 3464–3466. https://doi.org/10.1021/acs.est.3c01106.
Robert, M.H., Rosa Taghikhah, F., Nabavi, E., Razavi, S., Gupta, H., Wu, W., Radford, D. A.G., Huang, J., 2024. How much X is in XAI: responsible use of “Explainable” artificial intelligence in hydrology and water resources. J. Hydrol. X 25, 100185. https://doi.org/10.1016/j.hydroa.2024.100185.
Sakurada, M., Yairi, T., 2014. Anomaly detection using autoencoders with nonlinear dimensionality reduction. In: Proceedings of the MLSDA 2014 2nd Workshop on Machine Learning for Sensory Data Analysis, MLSDA’14. Association for Computing Machinery, New York, NY, USA, pp. 4–11. https://doi.org/10.1145/2689746.2689747.
Santos-Fernandez, E., Ver Hoef, J.M., Peterson, E.E., McGree, J., Villa, C.A., Leigh, C., Turner, R., Roberts, C., Mengersen, K., 2024. Unsupervised anomaly detection in spatio-temporal stream network sensor data. Water Resour. Res. 60. https://doi.org/10.1029/2023WR035707 e2023WR035707.
Sarthi, P., Abdullah, S., Tuli, A., Khanna, S., Goldie, A., Manning, C.D., 2023. RAPTOR: recursive abstractive processing for tree-organized retrieval. In: Presented at the Twelfth International Conference on Learning Representations. https://openreview.net/forum?id=GN921JHCRw&noteId=3W94b2afxu.
Schaefer, J.T., 1990. The critical success index as an indicator of warning skill. Weather Forecast. 5 (2), 570–575. https://doi.org/10.1175/1520-0434(1990)005%3C0570:TCSIAA%3E2.0.CO.
Schölkopf, B., Platt, J.C., Shawe-Taylor, J.C., Smola, A.J., Williamson, R.C., 2001. Estimating the support of a high-dimensional distribution. Neural Comput. 13, 1443–1471. https://doi.org/10.1162/089976601750264965.
Sela, L., Sowby, R.B., Salomons, E., Housh, M., 2025. Making waves: the potential of generative AI in water utility operations. Water Res. 272, 122935. https://doi.org/10.1016/j.watres.2024.122935.
Song, C., Zhang, H., 2020. Study on turbidity prediction method of reservoirs based on long short term memory neural network. Ecol. Model. 432, 109210. https://doi.org/10.1016/j.ecolmodel.2020.109210.
Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R., 2014. Dropout: a simple way to prevent neural networks from overfitting. J. Mach. Learn. Res. 15, 1929–1958. https://doi.org/10.5555/2627435.2670313.
Stets, E.G., Sprague, L.A., Oelsner, G.P., Johnson, H.M., Murphy, J.C., Ryberg, K., Vecchia, A.V., Zuellig, R.E., Falcone, J.A., Riskin, M.L., 2020. Landscape drivers of dynamic change in water quality of U.S. Rivers. Environ. Sci. Technol. 54, 4336–4343. https://doi.org/10.1021/acs.est.9b05344.
Storey, M.V., van der Gaag, B., Burns, B.P., 2011. Advances in on-line drinking water quality monitoring and early warning systems. Water Res. 45, 741–747. https://doi.org/10.1016/j.watres.2010.08.049.
Theocharidou, Marianthi, 2019. Practical Guidelines on Requirements of Continuous Online water-quality Monitoring System in drinking-water Supply Systems. Publications Office of the European Union. https://doi.org/10.2760/033873.
Tuli, S., Casale, G., Jennings, N.R., 2022. TranAD: deep transformer networks for anomaly detection in multivariate time series data. Proc. VLDB Endow. 15, 1201–1214. https://doi.org/10.14778/3514061.3514067.
US EPA, 2018. Drinking Water Standards and Health Advisories. https://www.epa.gov/system/files/documents/2022-01/dwtable2018.pdf.
US EPA, 2026a. 40 CFR part 141: other safe drinking water act regulations. Electronic Code of Federal Regulations (eCFR). https://www.ecfr.gov/current/title-40/chapter-I/subchapter-D/part-141.
US EPA, 2026b. 40 CFR part 143: other safe drinking water act regulations. In: Electronic Code of Federal Regulations (eCFR). Monitoring Compliance, Overviews and Factsheets. https://www.ecfr.gov/current/title-40/chapter-I/subchapter-D/part-143.
US EPA, 2026c. Monitoring Compliance, Overviews and Factsheets. https://www.epa.gov/compliance/monitoring-compliance.
US EPA, 2026d. Safe drinking water act (SDWA). https://www.epa.gov/sdwa.
US EPA, 2025. Safe Drinking Water Act. Drinking water regulations and contaminants. https://www.epa.gov/sdwa/drinking-water-regulations-and-contaminants.
USDA, 2025. Wildfire and water security research project | US forest service research and development. https://research.fs.usda.gov/pnw/projects/wildfire-water-security.
USGS, 2020. Responds to Oregon wildfires. U.S. Geological Survey. https://www.usgs.gov/news/usgs-responds-oregon-wildfires.
USGS, 2024. Bay-delta. U.S. Geological Survey. https://www.usgs.gov/centers/california-water-science-center/science/science-topics/bay-delta.
USGS, 2026a. USGS water data for the nation. https://waterdata.usgs.gov/.
USGS, 2026b. Potomac river near wash, DC little falls pump sta. http://waterdata.usgs.gov/monitoring-location/USGS-01646500/.
USGS, 2026c. White river near nora, IN. http://waterdata.usgs.gov/monitoring-location/USGS-03351000/.
USGS, 2026d. Clackamas river near Oregon city, OR. http://waterdata.usgs.gov/monitoring-location/USGS-14211010/.
USGS, 2026e. Sacramento river, freeport, CA. http://waterdata.usgs.gov/monitoring-location/USGS-11447650/.
USGS, 2026f. USGS water data APIs. https://api.waterdata.usgs.gov/.
USGS, 2026g. Code dictionary - usgs_parameter_codes. https://waterdata.usgs.gov/code-dictionary/.
Wagner, R.J., Jr, R.W.B., Oblinger, C.J., Smith, B.A., 2006. Guidelines and standard procedures for continuous water-quality monitors: station operation, record computation, and data reporting. Techniques and Methods 1–D3. https://doi.org/10.3133/tm1D3.
Wai, K.P., Chia, M.Y., Koo, C.H., Huang, Y.F., Chong, W.C., 2022. Applications of deep learning in water quality management: a state-of-the-art review. J. Hydrol. 613, 128332. https://doi.org/10.1016/j.jhydrol.2022.128332.
Wang, L., Dong, H., Cao, Y., Hou, D., Zhang, G., 2023. Real-time water quality detection based on fluctuation feature analysis with the LSTM model. J. Hydroinform. 25, 140–149. https://doi.org/10.2166/hydro.2023.127.
Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., Zhou, M., 2020. MiniLM: deep self-attention distillation for task-agnostic compression of pre-trained transformers. In: Proceedings of the 34th International Conference on Neural Information Processing Systems (NIPS'20). Curran Associates, Inc., pp. 5776–5788. https://dl.acm.org/doi/abs/10.5555/3495724.3496209
Wang, Y.-Q., Tian, W., Yang, H.-L., Song, Y.-P., Chen, J.-J., Xu, Q.-Y., Yin, W.-X., Ding, L.-Q., Li, X.-Q., Wang, H.-T., Wang, A.-J., Wang, H.-C., 2025. Knowledge embedding and interpretable machine learning optimize comprehensive benefits for water treatment. npj Clean Water 8, 81. https://doi.org/10.1038/s41545-025-00510-1.
WHO, 2022. Guidelines for drinking-water Quality: Fourth Edition Incorporating the First and Second Addenda. https://www.who.int/publications/i/item/9789240045064.
Xiong, G., Jin, Q., Lu, Z., Zhang, A., 2024. Benchmarking retrieval-augmented generation for medicine. In: Ku, L.-W., Martins, A., Srikumar, V. (Eds.), Findings of the Association for Computational Linguistics: ACL 2024. Presented at the Findings 2024. Association for Computational Linguistics, Bangkok, Thailand, pp. 6233–6251. https://doi.org/10.18653/v1/2024.findings-acl.372.
Xu, B., Wu, G., Li, Z., Xu, G., Zeng, H., Tong, R., Ng, H.Y., 2025. Towards domain-adapted large language models for water and wastewater management: methods, datasets and benchmarking. npj Clean Water 8, 82. https://doi.org/10.1038/s41545-025-00509-8.
Xu, J., Wu, H., Wang, J., Long, M., 2022. Anomaly transformer: time series anomaly detection with association discrepancy. Presented at the International Conference on Learning Representations (ICLR).. https://openreview.net/forum?id=LzQQ89U1qm.
Yang, D., Zeng, L., Rao, J., Zhang, Y., 2025. Knowing you don't know: learning when to continue search in multi-round RAG through self-practicing. In: Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR ’25. Association for Computing Machinery, New York, NY, USA, pp. 1305–1315. https://doi.org/10.1145/3726302.3730018.
Yaroshenko, I., Kirsanov, D., Marjanovic, M., Lieberzeit, P.A., Korostynska, O., Mason, A., Frau, I., Legin, A., 2020. Real-time water quality monitoring with chemical sensors. Sensors 20, 3432. https://doi.org/10.3390/s20123432.
Zamanzadeh Darban, Z., Webb, G.I., Pan, S., Aggarwal, C., Salehi, M., 2024. Deep learning for time series anomaly detection: a survey. ACM Comput. Surv. 57 (15), 1–42. https://doi.org/10.1145/3691338, 15.
Zeng, A., Chen, M., Zhang, L., Xu, Q., 2023. Are transformers effective for time series forecasting? Proc. AAAI Conf. Artif. Intell. 37, 11121–11128. https://doi.org/10.1609/aaai.v37i9.26317.
Zhao, J., Wei, S., Wen, X., Qiu, X., 2020. Analysis and prediction of big stream data in real-time water quality monitoring system. J. Ambient Intell. Smart Environ. 12, 393–406. https://doi.org/10.3233/AIS-200571.
Zheng, Y., Zhang, X., Zhou, Y., Zhang, Y., Zhang, T., Farmani, R., 2025. Deep representation learning enables cross-basin water quality prediction under data-scarce conditions. npj Clean Water 8, 33. https://doi.org/10.1038/s41545-025-00466-2.
Zhi, W., Appling, A.P., Golden, H.E., Podgorski, J., Li, L., 2024. Deep learning for water quality. Nat. Water 2, 228–241. https://doi.org/10.1038/s44221-024-00202-z.
Zhi, W., Feng, D., Tsai, W.-P., Sterle, G., Harpold, A., Shen, C., Li, L., 2021. From hydrometeorology to river water quality: can a deep learning model predict

dissolved oxygen at the Continental scale? Environ. Sci. Technol. 55, 2357–2368. https://doi.org/10.1021/acs.est.0c06783.

Zhong, X., Jin, B., Ouyang, S., Shen, Y., Jin, Q., Fang, Y., Lu, Z., Han, J., 2025. Benchmarking retrieval-augmented generation for chemistry. Presented at the Second Conference on Language Modeling, Second Conference on Language Modeling.. https://openreview.net/forum?id=GPu0orCDfN.

Zhu, J.-J., Yang, M., Jiang, J., Bai, Y., Chen, D., Ren, Z.J., 2024. Enabling GPTs for expert-level environmental engineering question answering. Environ. Sci. Technol. Lett. 11, 1327–1333. https://doi.org/10.1021/acs.estlett.4c00665.